

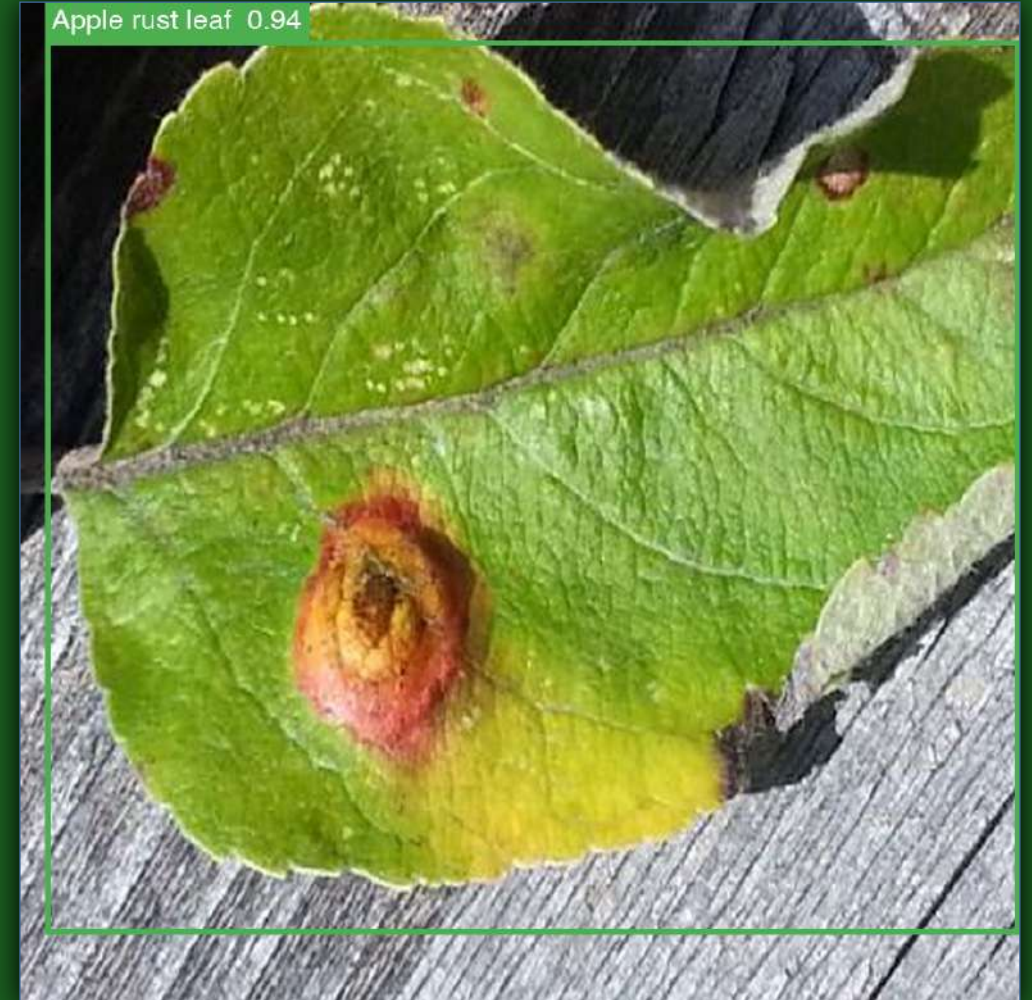
DATA MINING AND MACHINE LEARNING'S PROJECT

*Università degli Studi di Napoli Federico II*

# ***Plant Disease Detection***

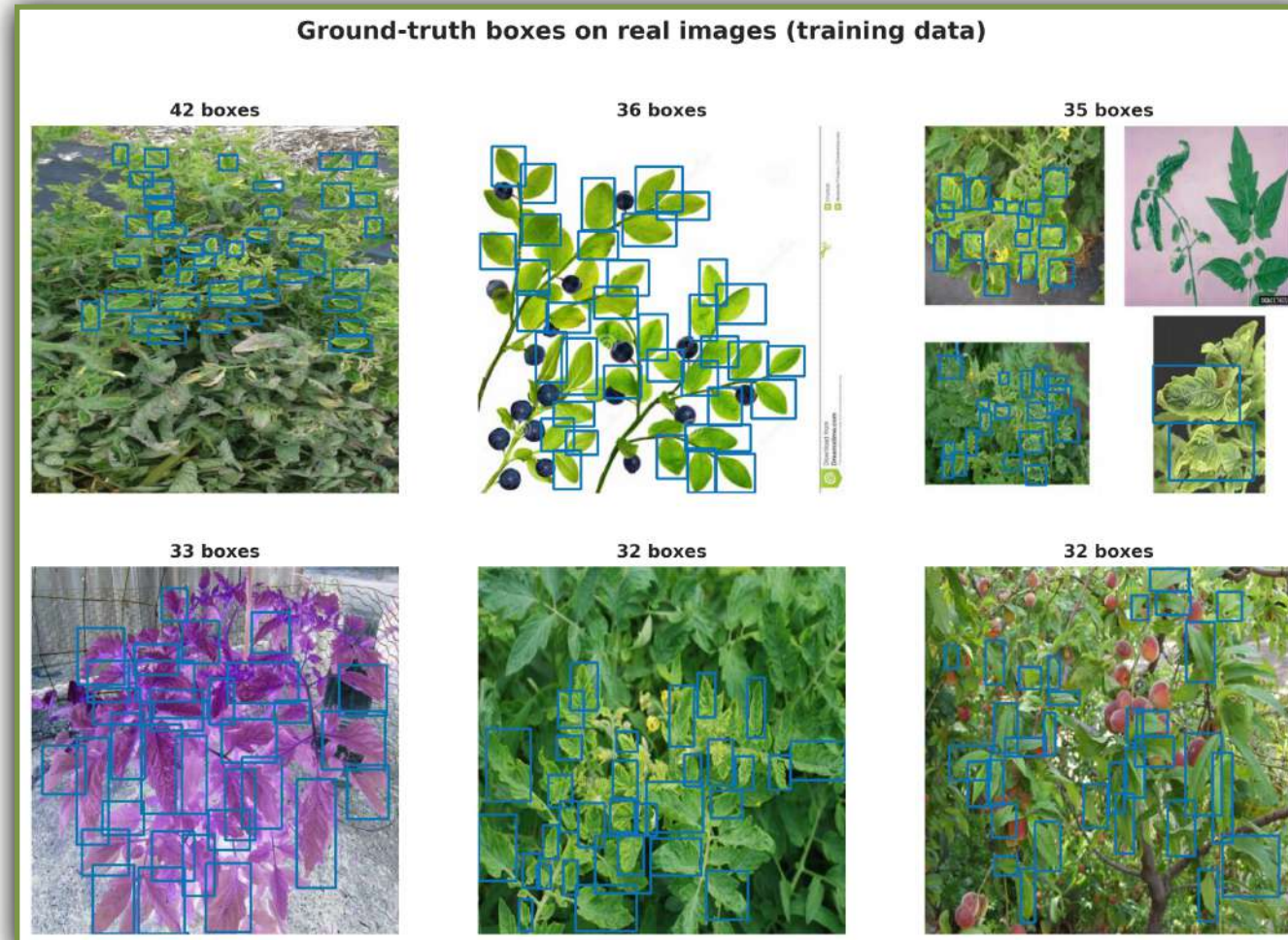
*Transfer learning on PlantDoc,  
By using YOLOv8s*

Diako Darabi



*Apple rust leaf · YOLOv8s detection*

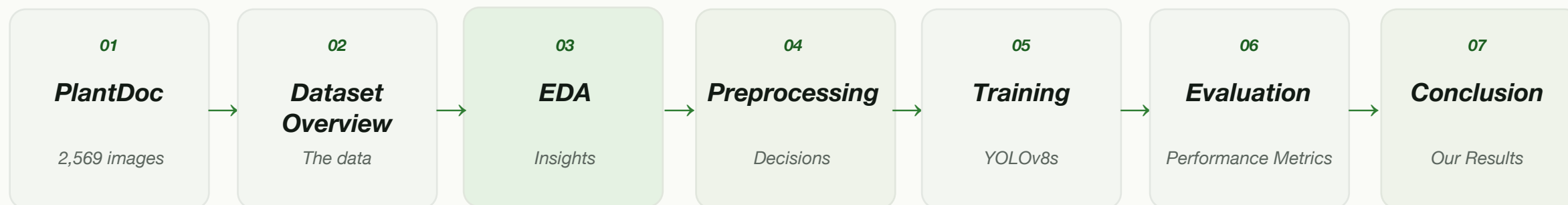
# Detection vs. Classification



Every leaf must be **localize** and **classify**

# Pipeline

The task: *Object Detection*



*A Data-Driven Study on the PlantDoc Dataset*

ACT II

# *Meeting the Data*

*what we are working with.*

---





# PlantDoc — 416x416

Naturalistic — Public data set published by **Roboflow** — Small in **JPG format** associated with one **text file**

**2,567**

IMAGES

**30**

CLASSES

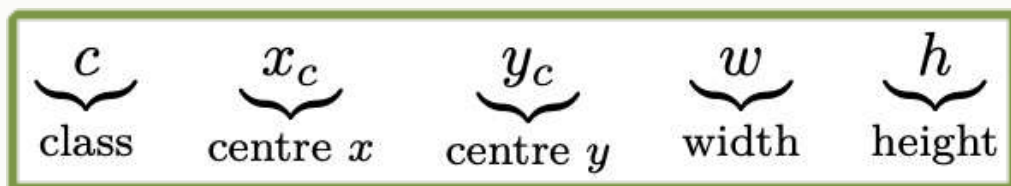
**8,807**

ANNOTATIONS

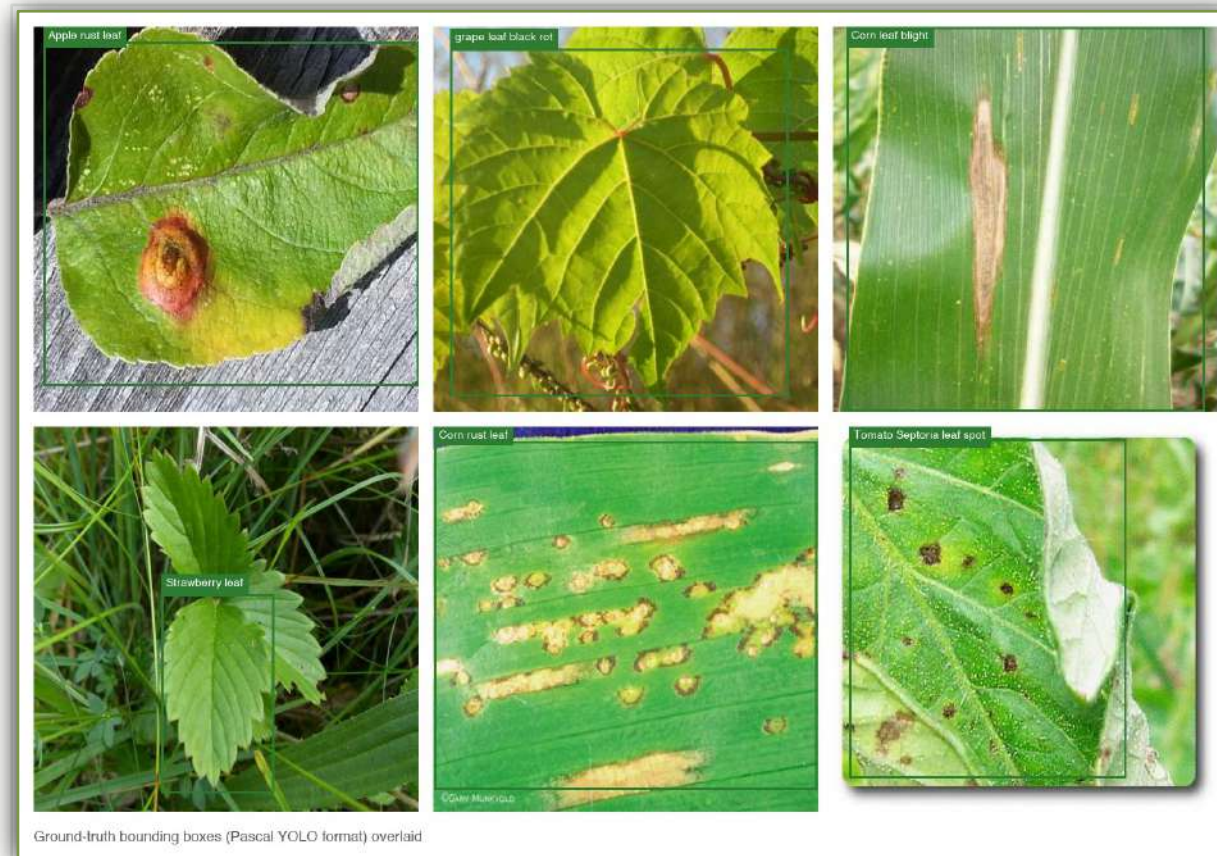
What is Annotation ?

# Annotation

8,807 ground-truth boxes — and what they actually look like in the data.



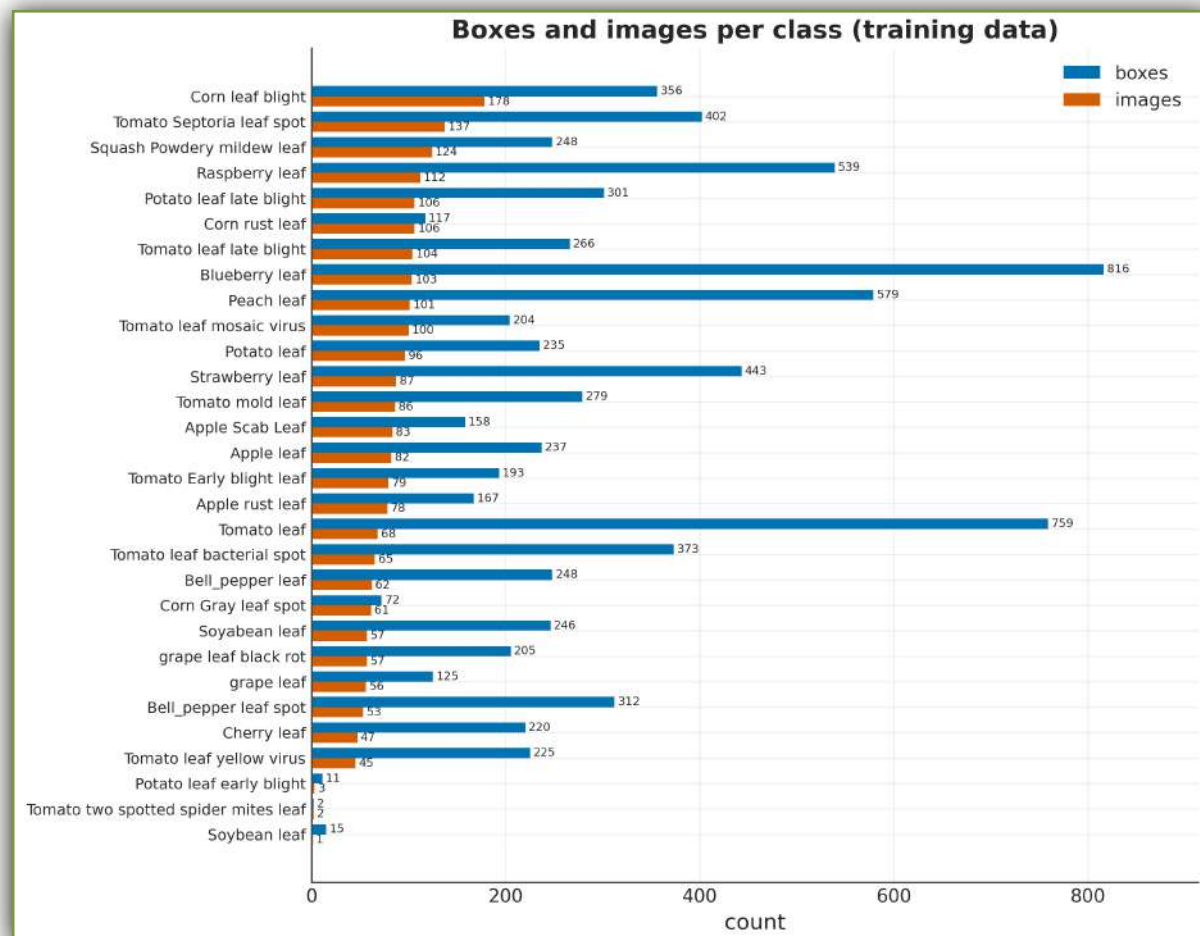
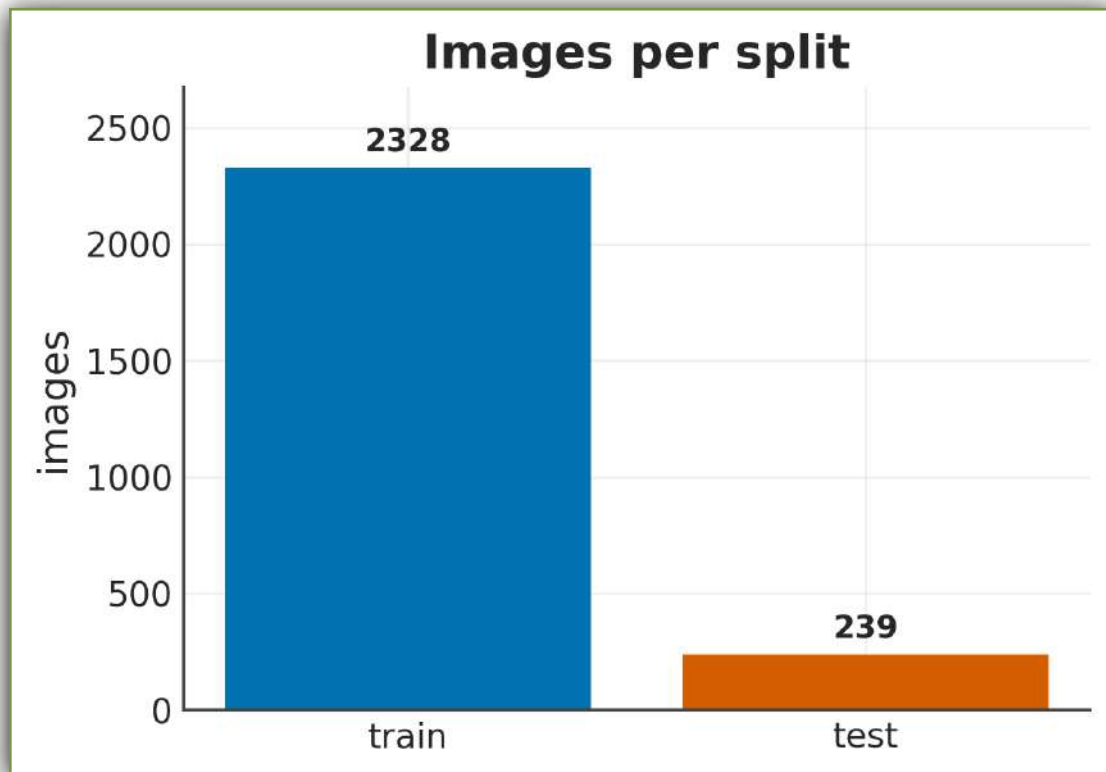
|                | Train | Test | Total |
|----------------|-------|------|-------|
| Images         | 2,328 | 239  | 2,567 |
| Bounding boxes | 8,353 | 454  | 8,807 |



*Normalized coordinates stay correct at any display size as fractions of the image (0–1)*

# Images per-split and what classes we have

*Dataset imbalance exists within training model may underperform on rare Class*



*No validation set and the test set is small*

ACT III

# *Exploratory Data Analysis*

*Each finding becomes a modelling decision or a documented limitation.*

---





# Step 0 — can we trust the data?

Three strict checks

## *Range check*

***every class number must be in the allowed list***

***every box corner inside the image***

***every box wider and taller than zero***

## *Readability check*

***every image must actually open.***

## *Pairing check*

***every image must have exactly one label  
file***

# Step 1 — The leakage trap, Is the Split is Really Clean?

Perceptual-hash audit on every image pair across splits.

Train/test leakage audit — 2 of 239 test images are content-identical (pHash 0) to a train image

test image



matched train image (Hamming 0)



test image



matched train image (Hamming 0)



*Duplicate pairs*

**2**

*Test images contaminated*

**0.84%**

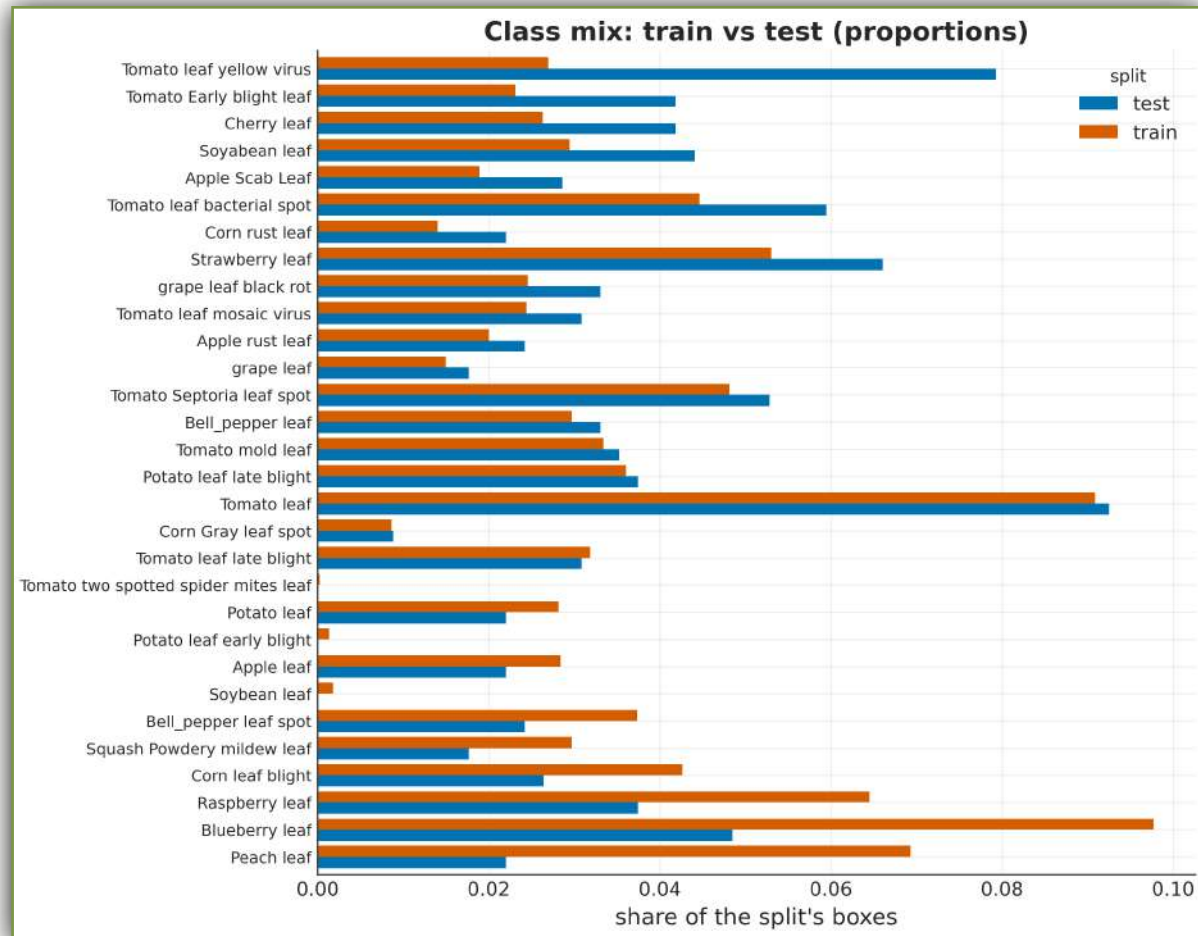
(2 of 239)

*Near duplicates*

**0**

## Step 2 — What classes are in the data ?

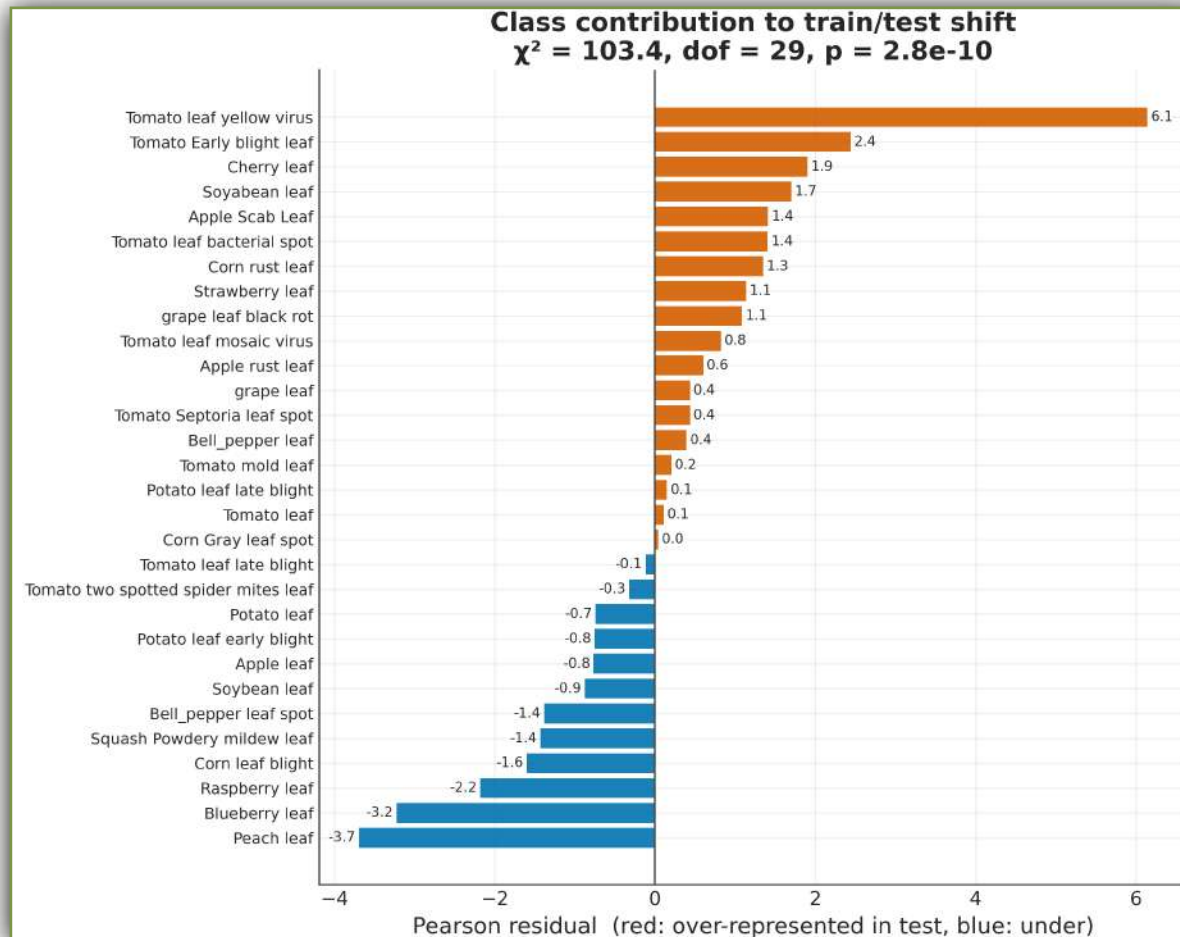
This Graph shows *potential train-test class distribution shift*



*We have asses this statistically as well !*

## Step 3 — do the train and test set match?

Explains why some classes may have higher or lower AP due to **test composition differences**.



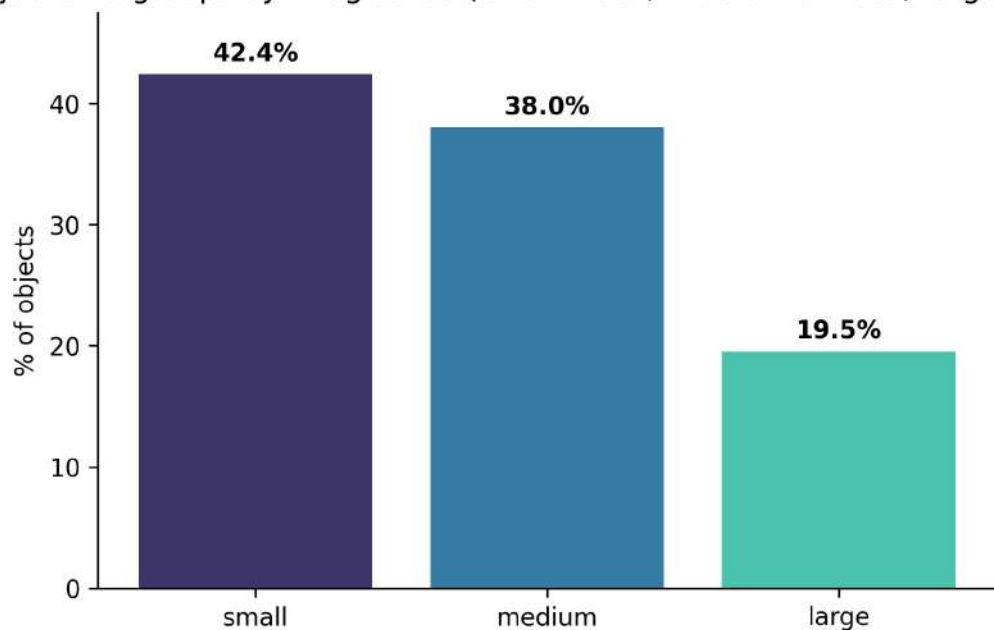
**Test class distribution may affect the model performance, not learning quality.**



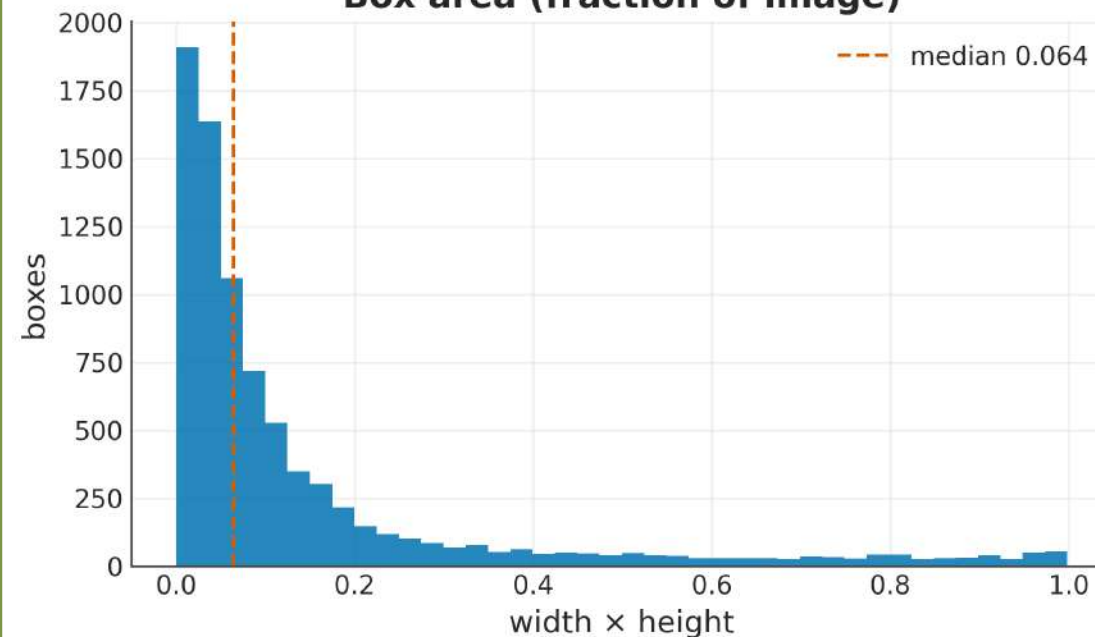
## Step 4 — How big are the Bounding boxes?

*Almost half* of all leaves are tiny → difficult for YOLO to detect reliably

Object size groups by image area (small <5% / medium 5-20% / large >=20%)



Box area (fraction of image)



*Small leaves are harder to detect because they contain fewer pixels and less texture*

## Small Objects in the Dataset: Real Examples and Pixel Coverage

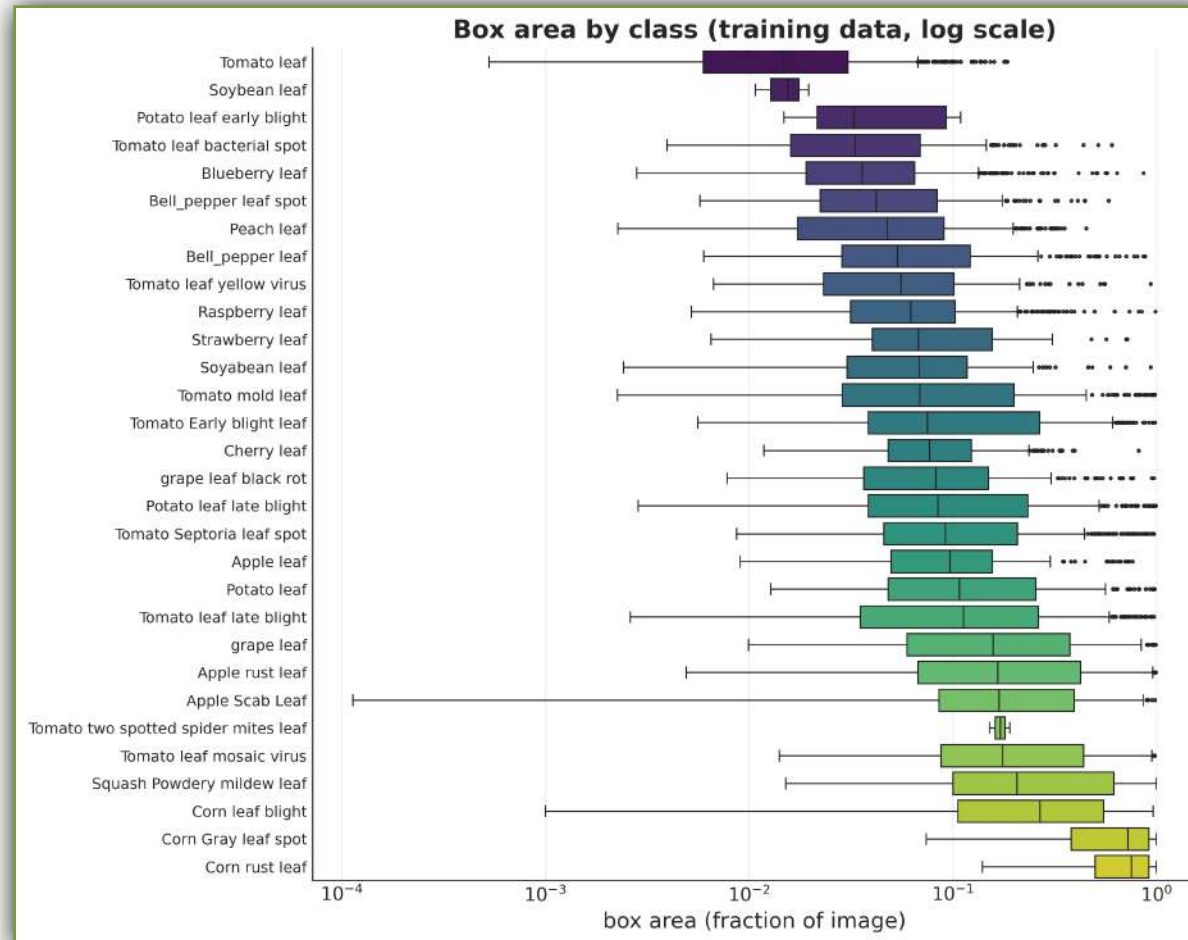
*Tiny objects are real and they drive the input size decision.*



*We resized images to 640 px so even the smallest leaves are large enough for the model to detect.*

## Step 5 — Which classes carry the smallest objects?

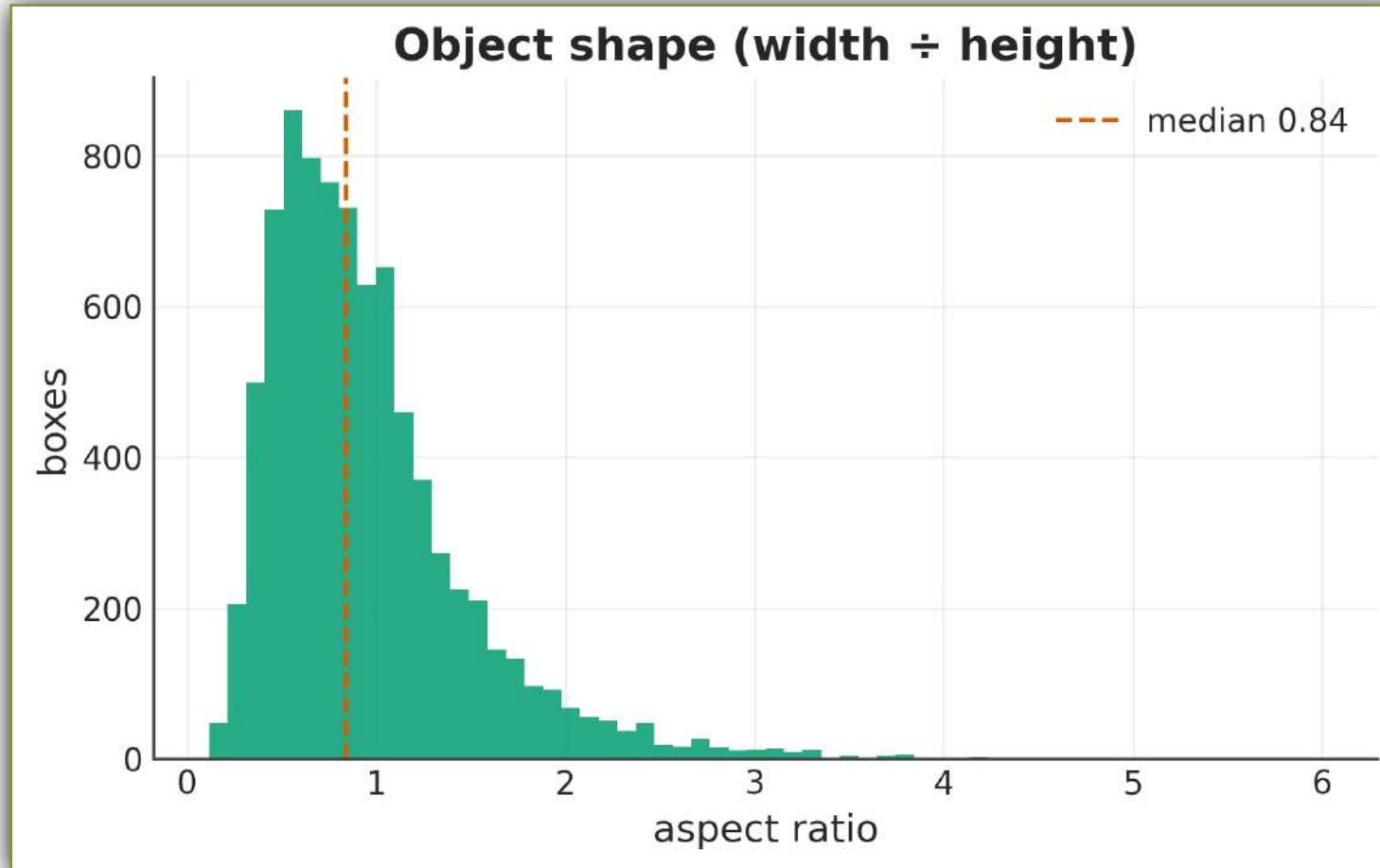
*Per-class spread of box area (log scale), ordered by median.*



**Object size isn't uniform across classes.**

## Step 6 — What shape are the Bounding Boxes?

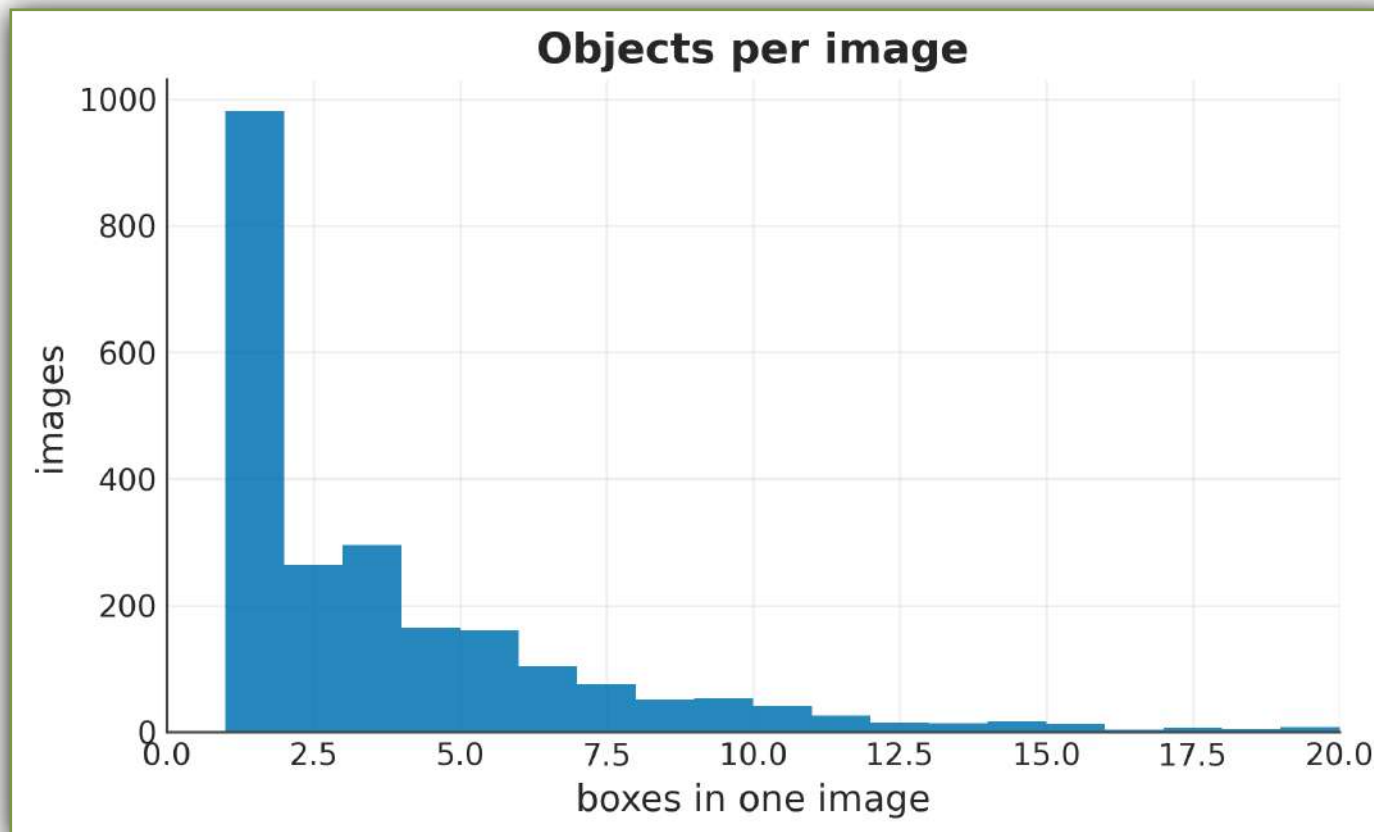
Bounding-box aspect ratio (width / height).



**Median 0.84 → boxes are mostly near-square**

## Step 7 — how busy are the Images?

Images are **mostly sparse** and contain few objects, which suggests potential augmentation settings.

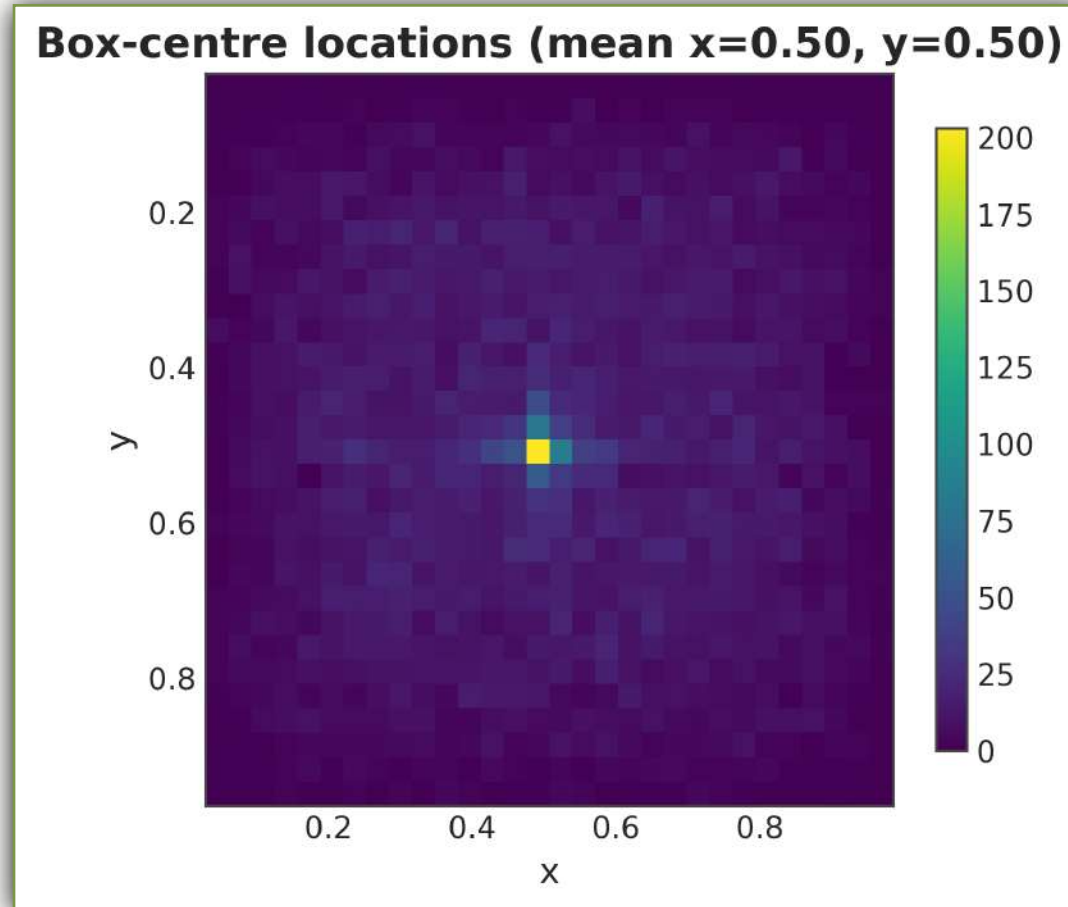


*Mosaic augmentation is safe here without overcrowding*



## Step 8 — where do objects sit?

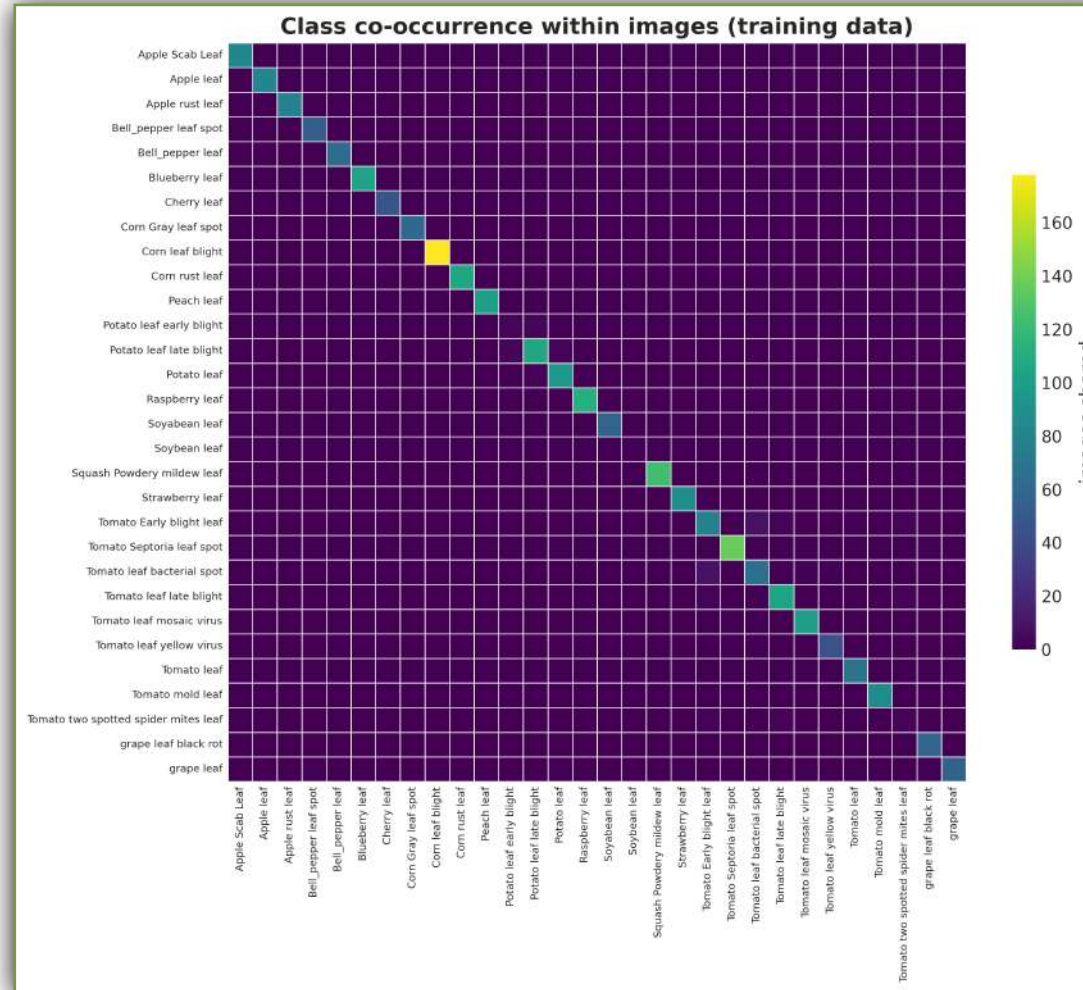
Objects are roughly centred, and they are *well spread*.



*The boxes are distributed symmetrically → **horizontal flip** can be safe for data augmentation*

## Step 9 — how many classes per Image?

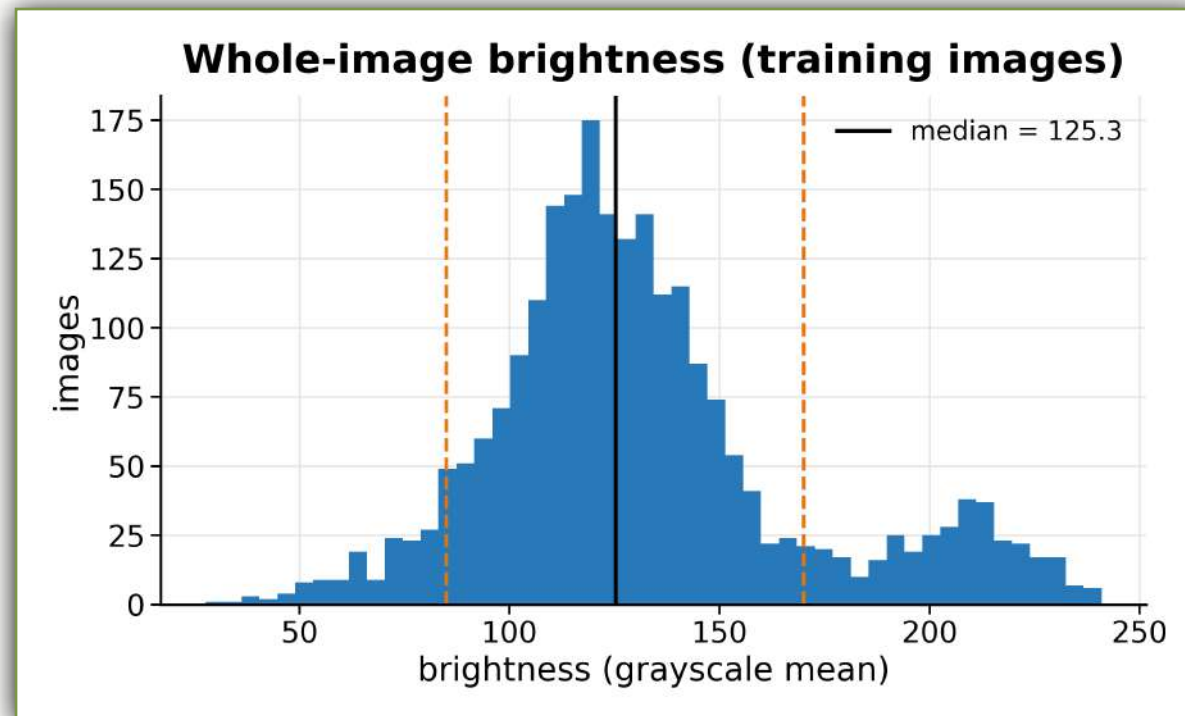
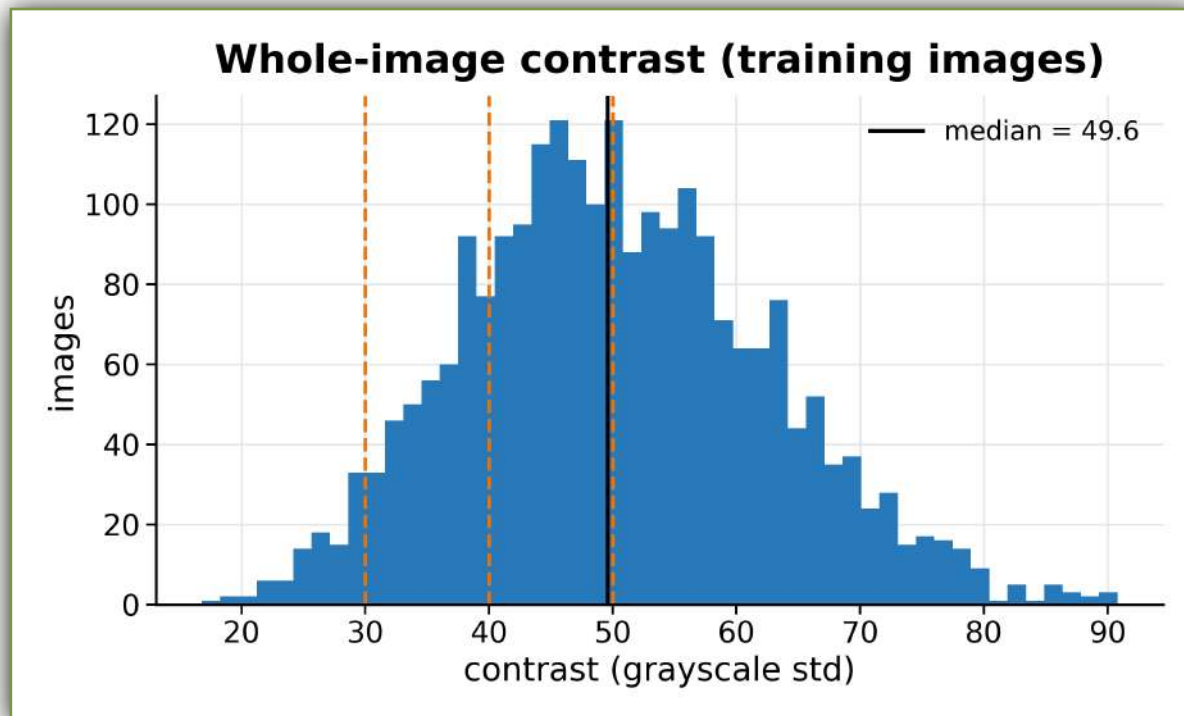
Images almost always contain a **single disease class**



**Low multi-class confusion expected due to dataset structure.**

## Step 10 — Is the image quality good ?

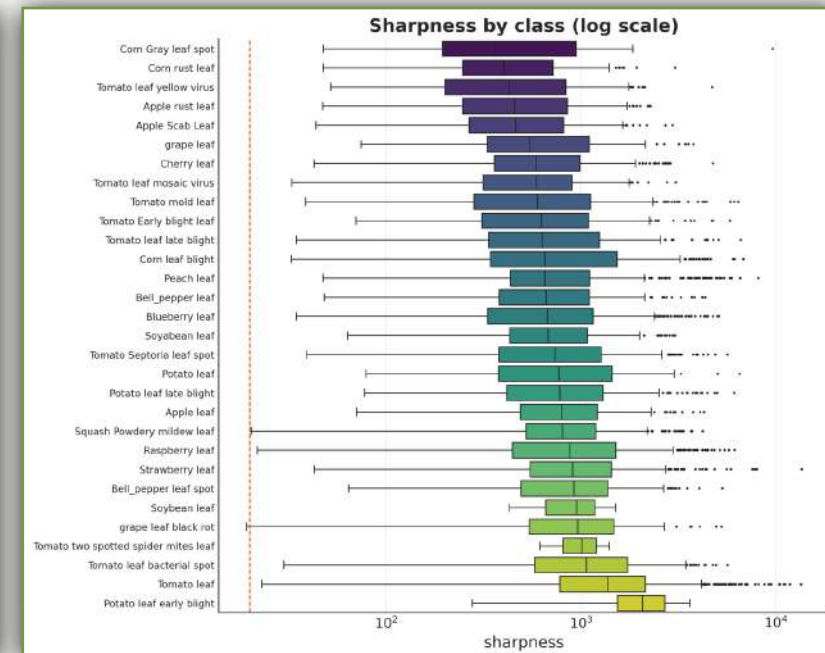
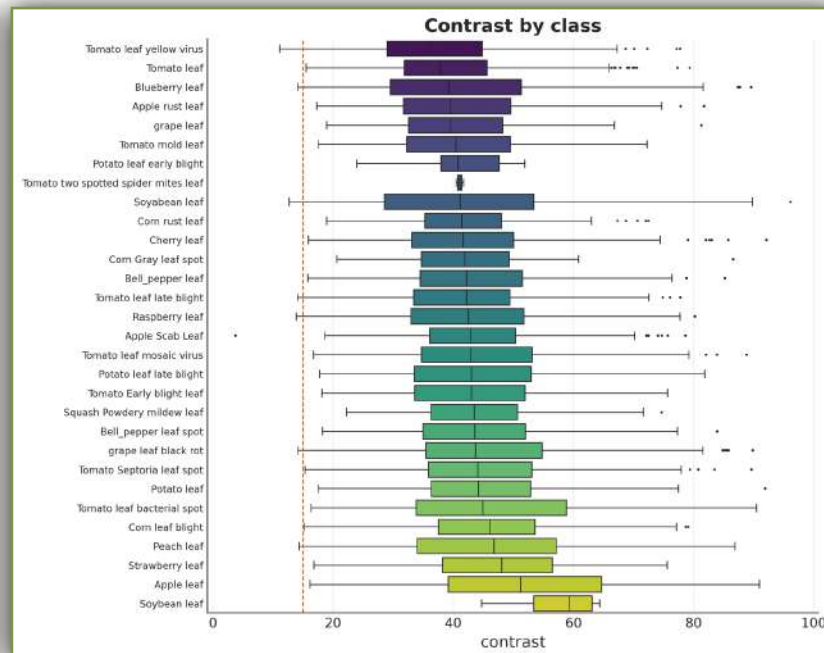
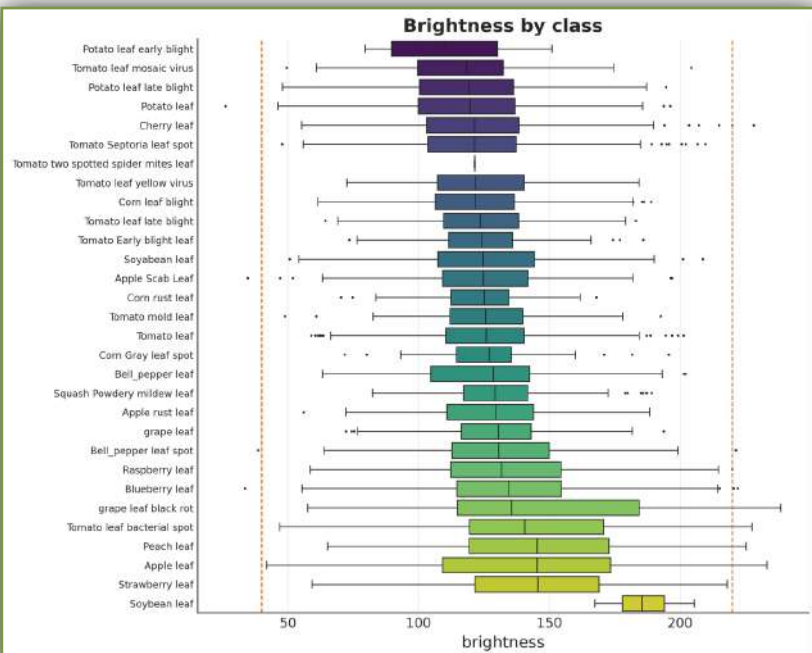
The dataset is *largely visually informative*



*Brightness and contrast augmentation can simulate realistic variation*

# Is the image quality good per class?

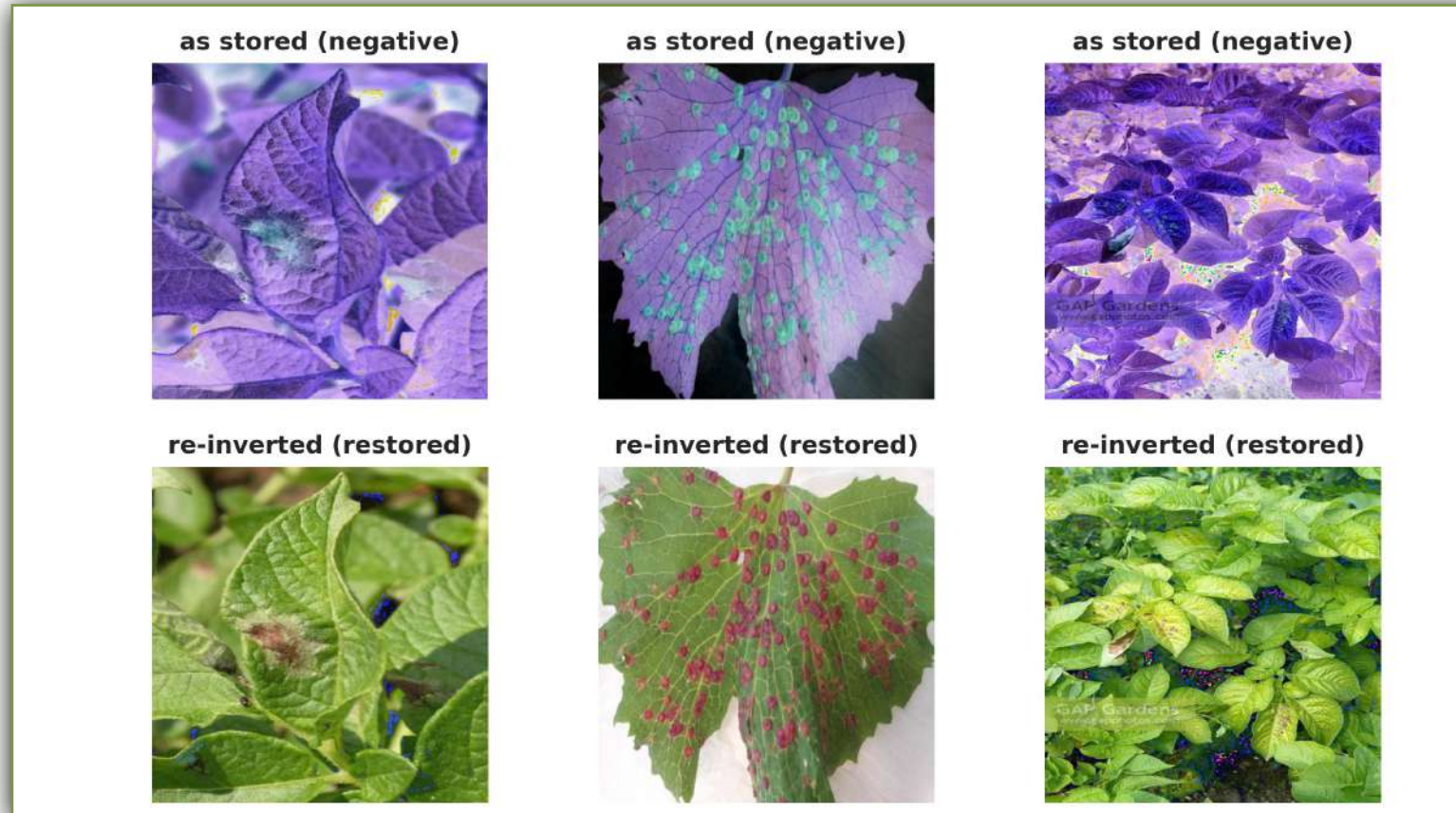
We inspect four things: *brightness*, *contrast*, *sharpness*



Confirms all classes have good image quality → no preprocessing or filtering needed

## Step 10 — THE TWIST: photo- negatives

The model could learn *colour-based shortcuts* (purple leaves → disease) instead of *shape/texture*



**22.6% of train and 18.0% of test images are colour negatives, we have to ensure** Ensures colour consistency across dataset



# Step 12 – Which Classes Are Visually Similar in Colour?

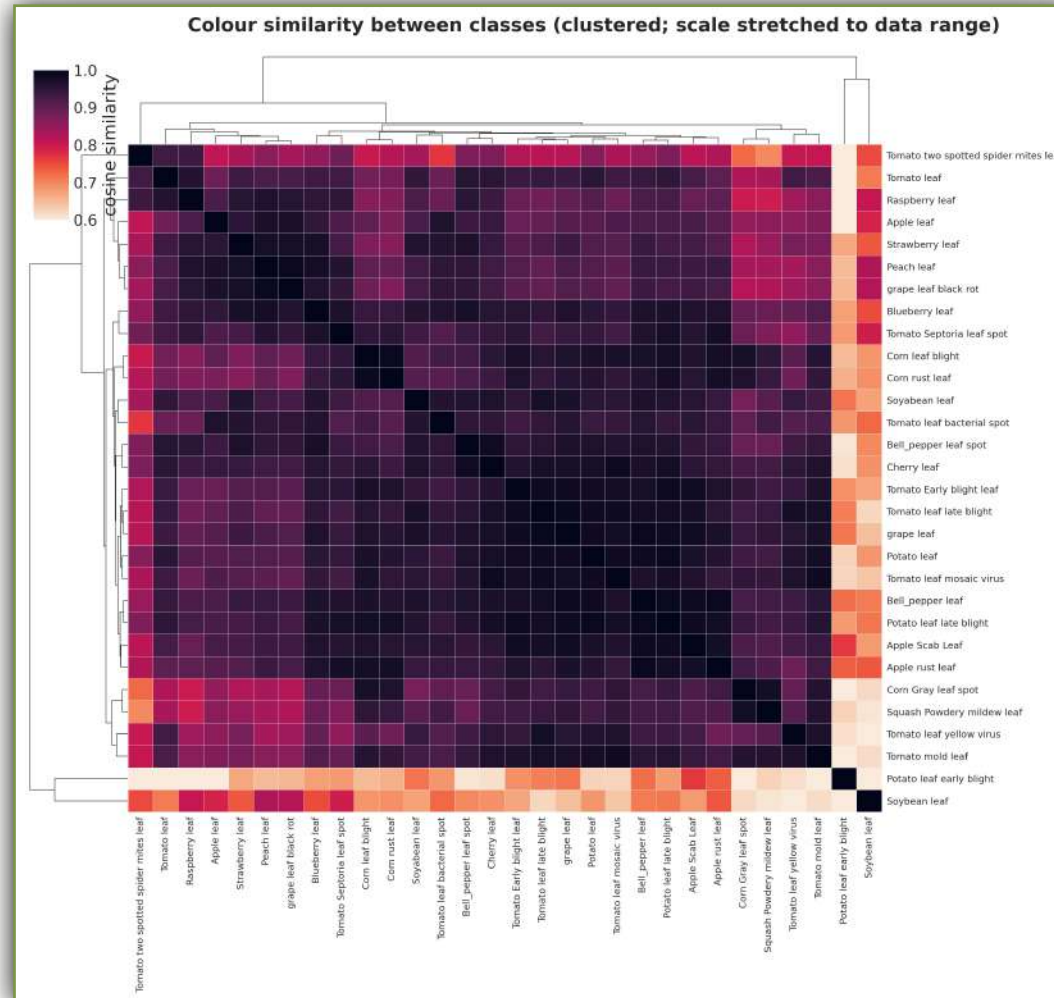
Average colour per class summarises the typical hue of each disease

| Average color by class               |     |     |     |
|--------------------------------------|-----|-----|-----|
|                                      | R   | G   | B   |
| Apple Scab Leaf                      | 121 | 128 | 114 |
| Apple leaf                           | 131 | 149 | 115 |
| Apple rust leaf                      | 129 | 130 | 118 |
| Bell_pepper leaf                     | 124 | 130 | 109 |
| Bell_pepper leaf spot                | 124 | 143 | 94  |
| Blueberry leaf                       | 138 | 140 | 107 |
| Cherry leaf                          | 115 | 131 | 96  |
| Corn Gray leaf spot                  | 123 | 132 | 113 |
| Corn leaf blight                     | 119 | 125 | 110 |
| Corn rust leaf                       | 125 | 126 | 108 |
| Peach leaf                           | 141 | 155 | 114 |
| Potato leaf                          | 111 | 126 | 97  |
| Potato leaf early blight             | 123 | 93  | 173 |
| Potato leaf late blight              | 116 | 124 | 104 |
| Raspberry leaf                       | 128 | 146 | 89  |
| Soyabean leaf                        | 122 | 132 | 110 |
| Soybean leaf                         | 186 | 193 | 151 |
| Squash Powdery mildew leaf           | 126 | 134 | 121 |
| Strawberry leaf                      | 136 | 155 | 108 |
| Tomato Early blight leaf             | 114 | 132 | 112 |
| Tomato Septoria leaf spot            | 121 | 126 | 106 |
| Tomato leaf                          | 114 | 139 | 87  |
| Tomato leaf bacterial spot           | 138 | 151 | 127 |
| Tomato leaf late blight              | 116 | 130 | 108 |
| Tomato leaf mosaic virus             | 106 | 126 | 103 |
| Tomato leaf yellow virus             | 110 | 136 | 98  |
| Tomato mold leaf                     | 117 | 134 | 108 |
| Tomato two spotted spider mites leaf | 114 | 137 | 63  |
| grape leaf                           | 123 | 135 | 111 |
| grape leaf black rot                 | 138 | 155 | 115 |

Several classes have **similar average colours** → A deeper analysis is needed !

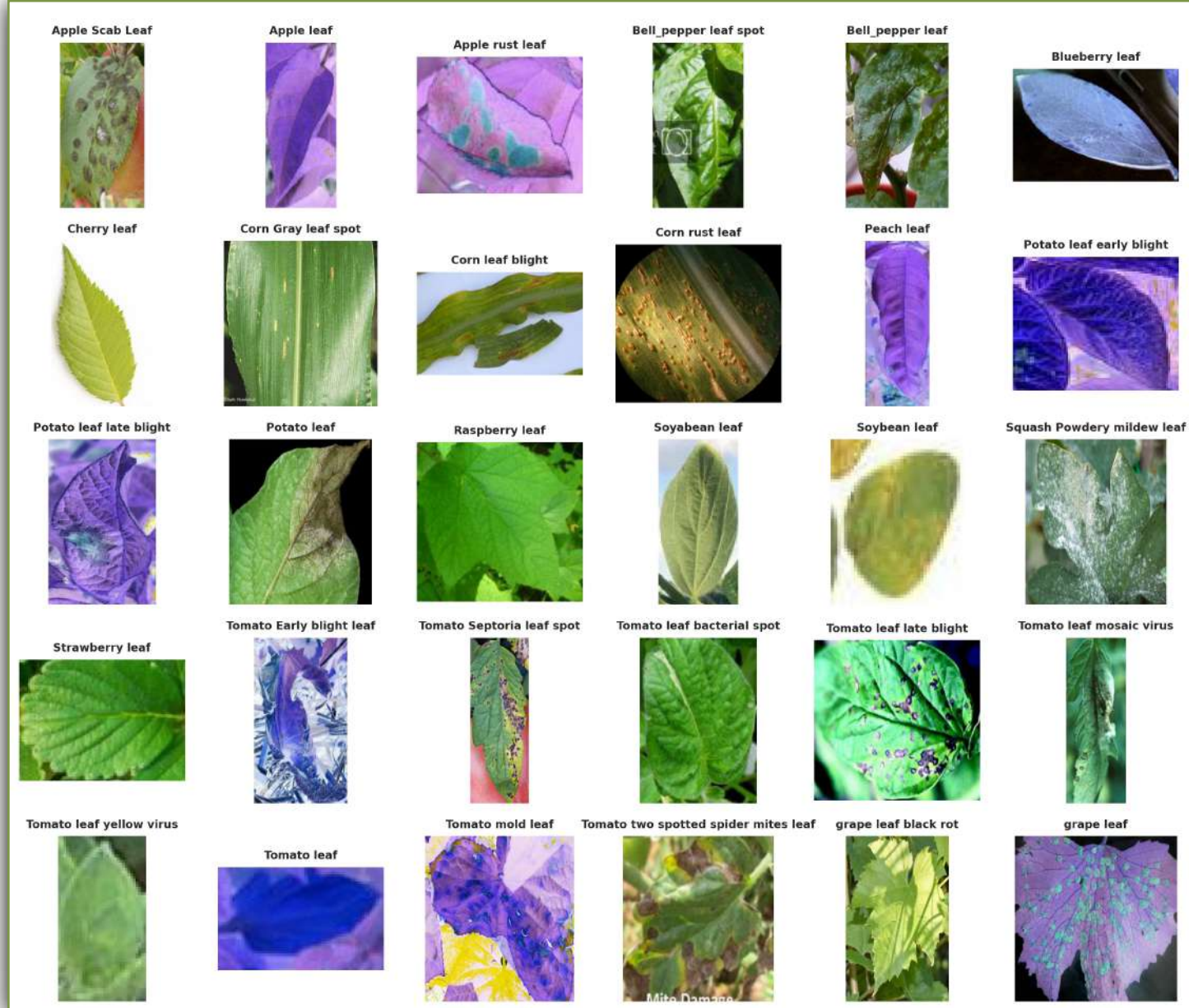
# Step 13 — colour similarity and clustering

30×30 cosine-similarity matrix on per-class colour histograms.



Several classes have really **similar colours** → Model need to focus on **Texture, shape and patterns** for classification

# Lets see the data with our own eyes





# boxes on real images

Ground-truth boxes drawn on the busiest photos.

Ground-truth boxes on real images (training data)

42 boxes



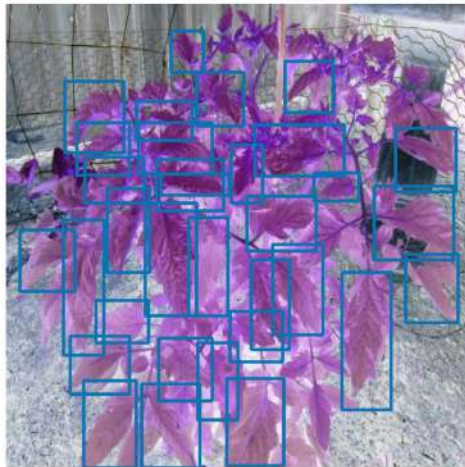
36 boxes



35 boxes



33 boxes



32 boxes



32 boxes



ACT IV

# *Preparing phase*

*Architecture · init · augmentation · class Imbalance. All four moved together.*

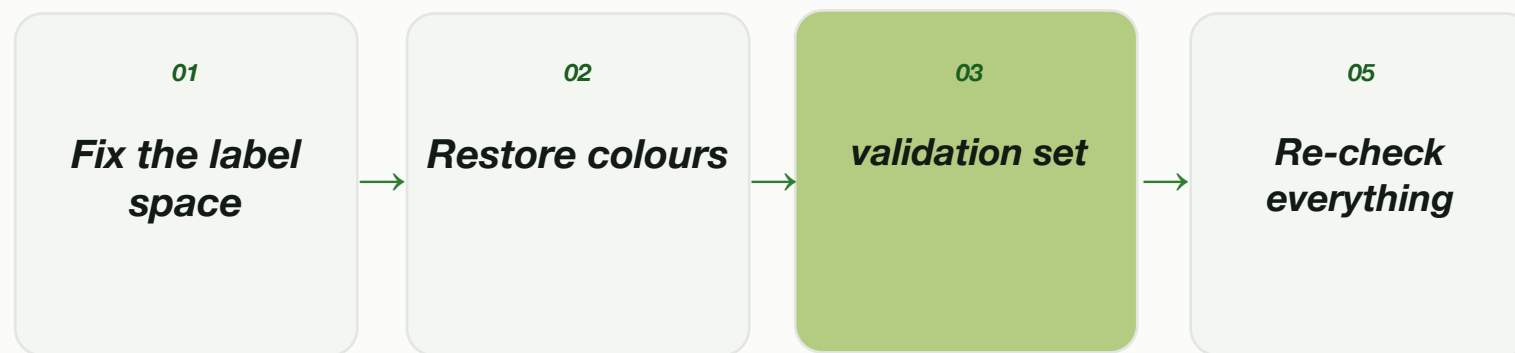
---





# Preprocessing Steps

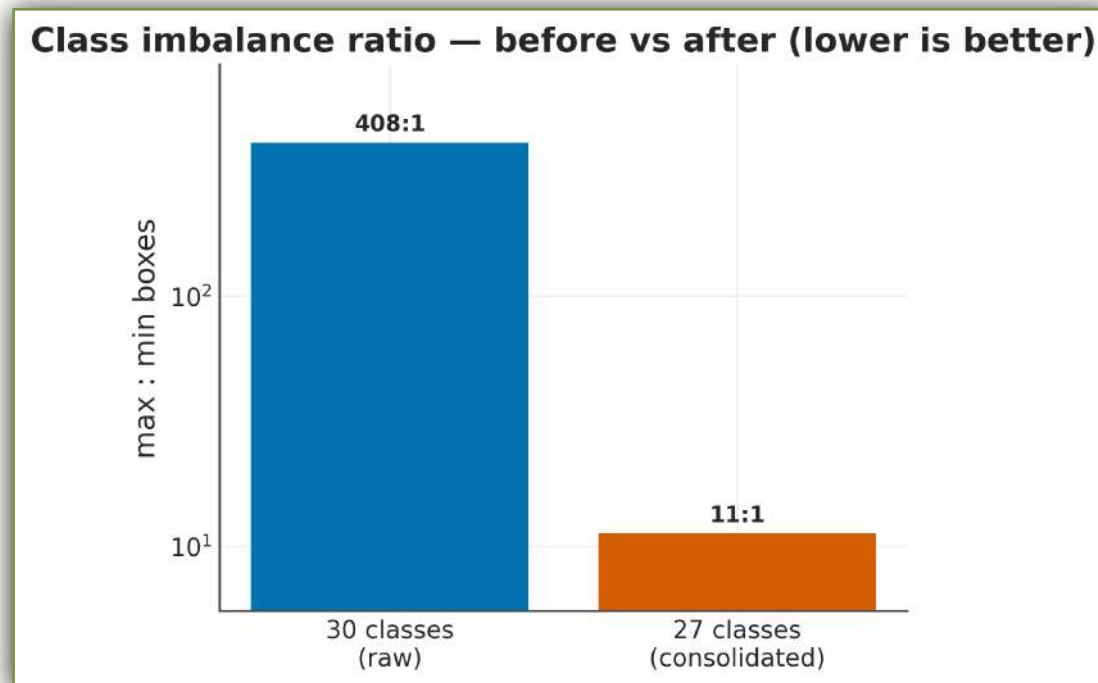
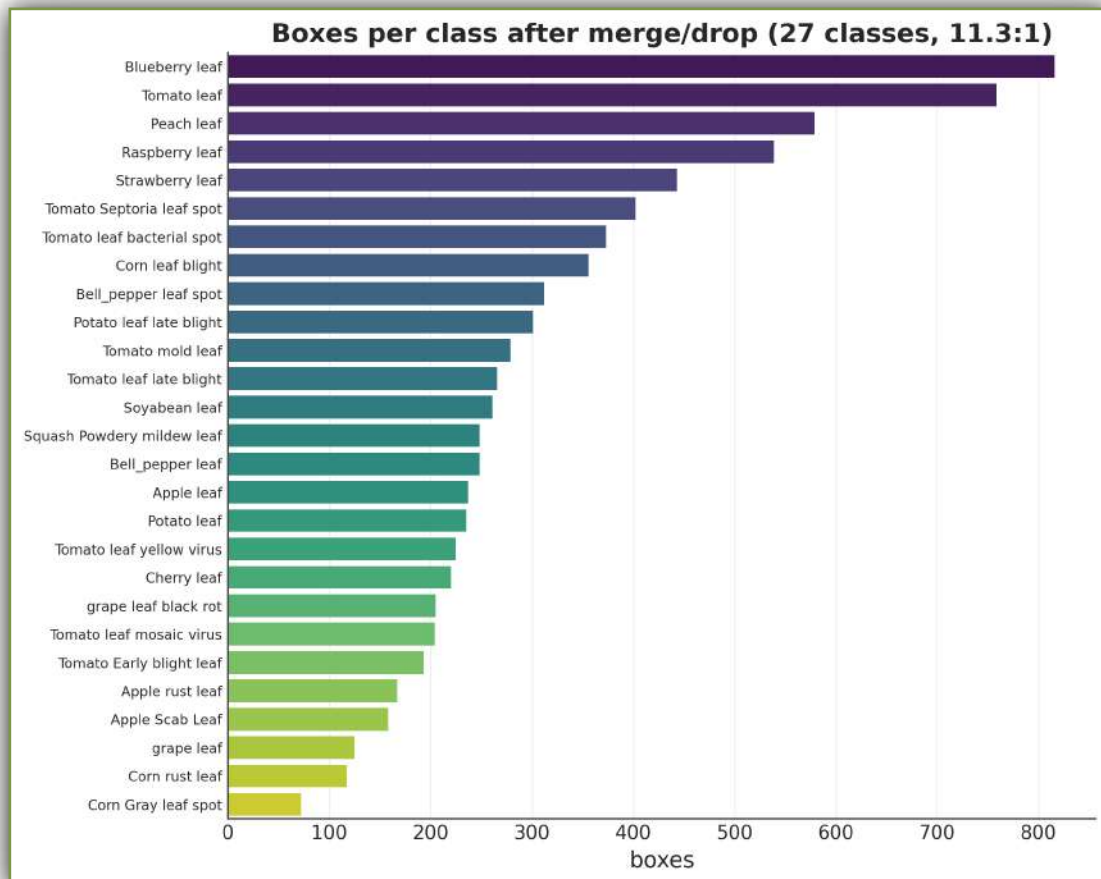
The preprocessing pipeline resizes all images to **640×640**



*Every step states its reason inline, and the steps run in a **leakage-safe** order.*

# Step 1 — From 30 classes to 27

We **merged** the *Soybean/Soyabeen* typo and **drop** the two un-learnable classes *Potato leaf early blight* — *Tomato two spotted spider mites leaf*



*One tidy step, turns a brutal imbalance into a manageable one — and every surviving class now has test data.*

## Step 2 — Restore the true colours

*It restores reality rather than improving it*

**Photo-negative defect — 22.6% of train, 18.0% of test (repaired in Phase 3 on all splits)**

as stored (negative)



as stored (negative)



as stored (negative)



re-inverted (restored)



re-inverted (restored)



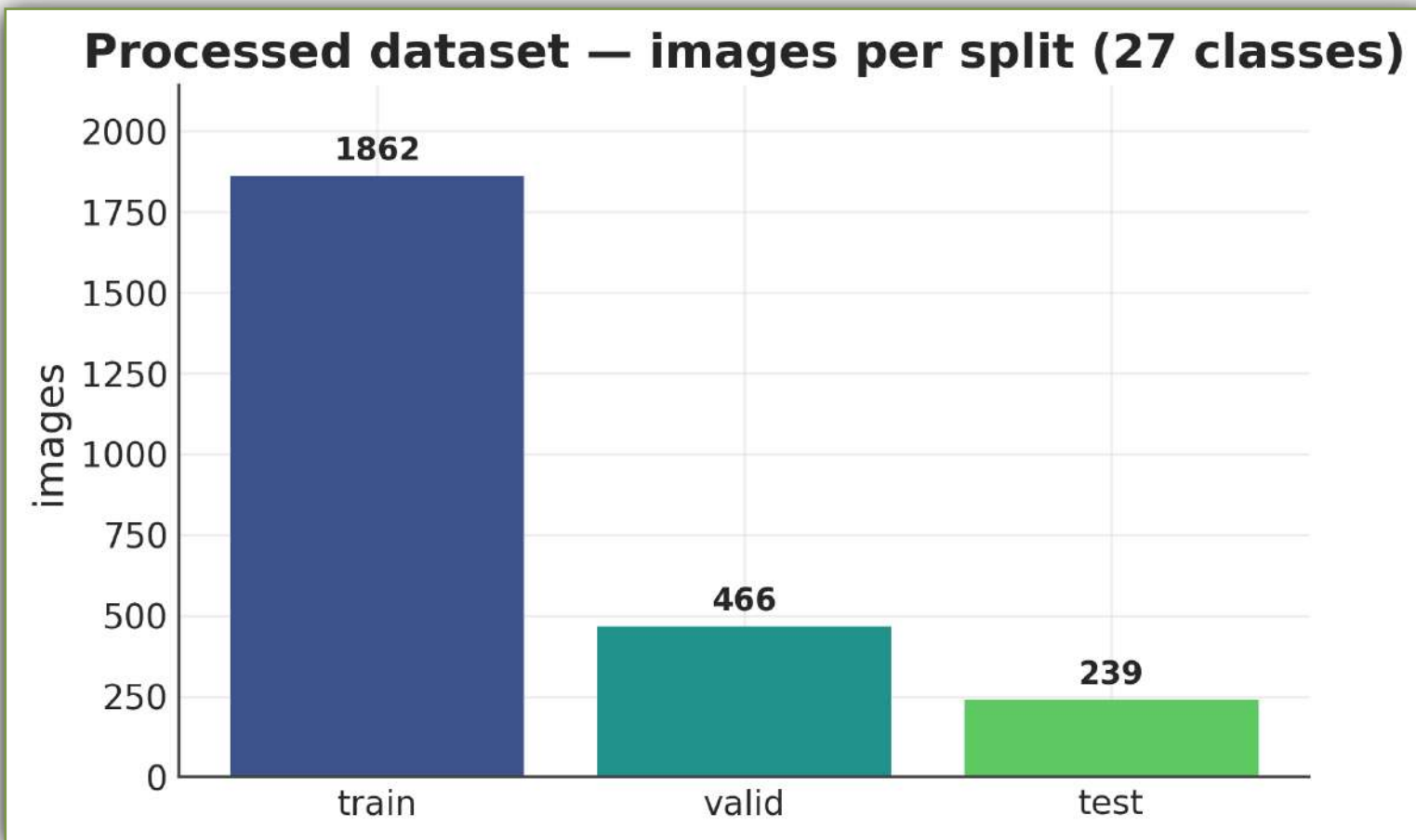
re-inverted (restored)



***The Only change to the Test set !***

## Step 3 — build the validation set, after cleaning

Hold back **20%** of training, split **by image** and **stratified** (each class's share preserved).



| Split      | Images | Boxes |
|------------|--------|-------|
| Train      | 1,862  | 6,568 |
| Validation | 466    | 1,772 |
| Test       | 239    | 454   |

**Result: 1,862 train — 466 validation.** By image, so all leaves of one photo stay together, stratified, so rare classes appear on both sides.



ACT V

# *Training*

*YOLOv8s from COCO weights · input 640 · batch 16 · up to 100 epochs · seed 42,  
deterministic.*

---



# YOLOv8 Training Pipeline

*Transfer learning, because the dataset is small.*

| Setting    | Value              | Setting                  | Value           |
|------------|--------------------|--------------------------|-----------------|
| Model      | YOLOv8s (COCO)     | Mosaic                   | 1.0             |
| Input size | 640 px             | Close mosaic             | last 10 epochs  |
| Batch size | 16                 | Horizontal flip          | 0.5             |
| Epochs     | up to 100          | HSV (h, s, v)            | 0.015, 0.7, 0.4 |
| Patience   | 20                 | Translate / Scale        | 0.1 / 0.5       |
| Seed       | 42 (deterministic) | Vertical flip / rotation | 0 / 0°          |

*The model rarely sees the **raw image** — each epoch it sees slightly modified versions, **improving generalisation**.*

# The augmentations settings which model will see

Improving **Generalisation** and preventing **Memorisation**

Train-time augmentations (EDA-justified): flip — object centres are symmetric · HSV jitter — per-class colour spread · mosaic — multi-object scenes

original (cleaned)



horizontal flip (p=0.5)



HSV jitter



translate 0.1 + scale 0.5



mosaic (4 images)



*Creating slightly different versions of each image every epoch to improve generalisation*

## ***Three progressive runs***

*The progressive approach shows an incremental improvement pipeline*

**Run 1** — Transfer baseline

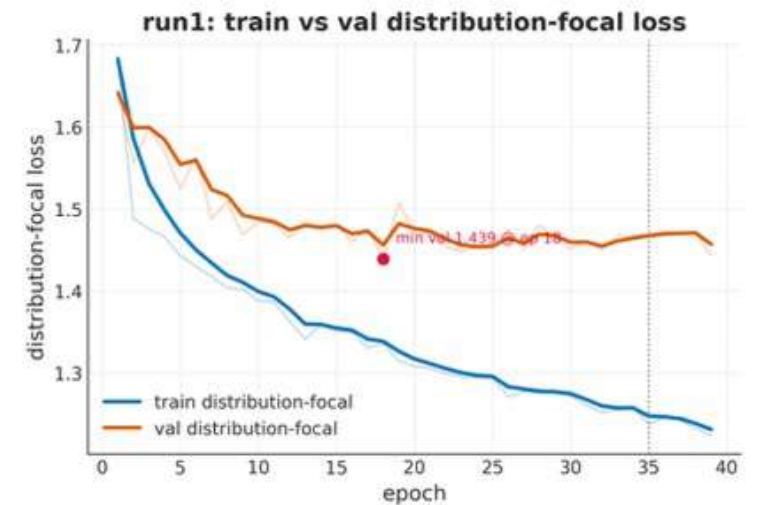
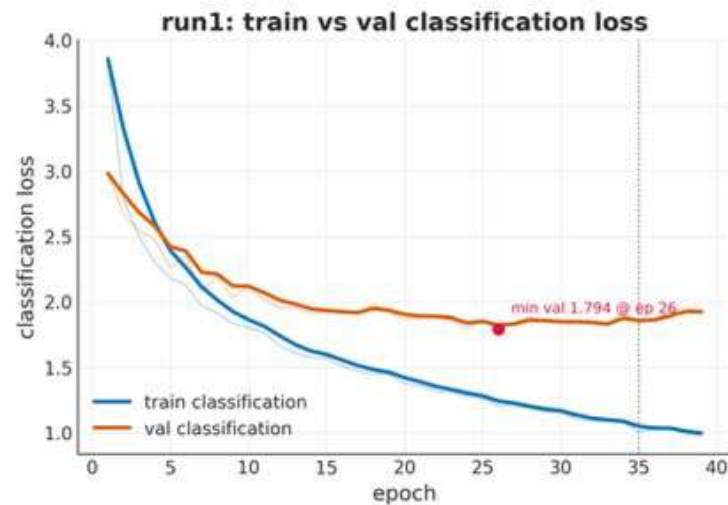
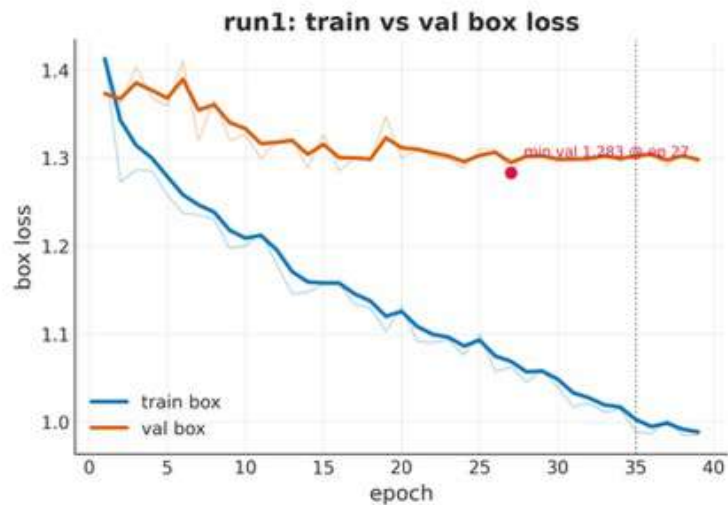
**Run 2** — AdamW + cosine

**Run 3** — Fine-tune + TTA

*All these models will be compared on validation set; the winner is scored once on test.*

# Run 1 – Training Loss

*A clean reference point: transfer learning alone on the cleaned data.*

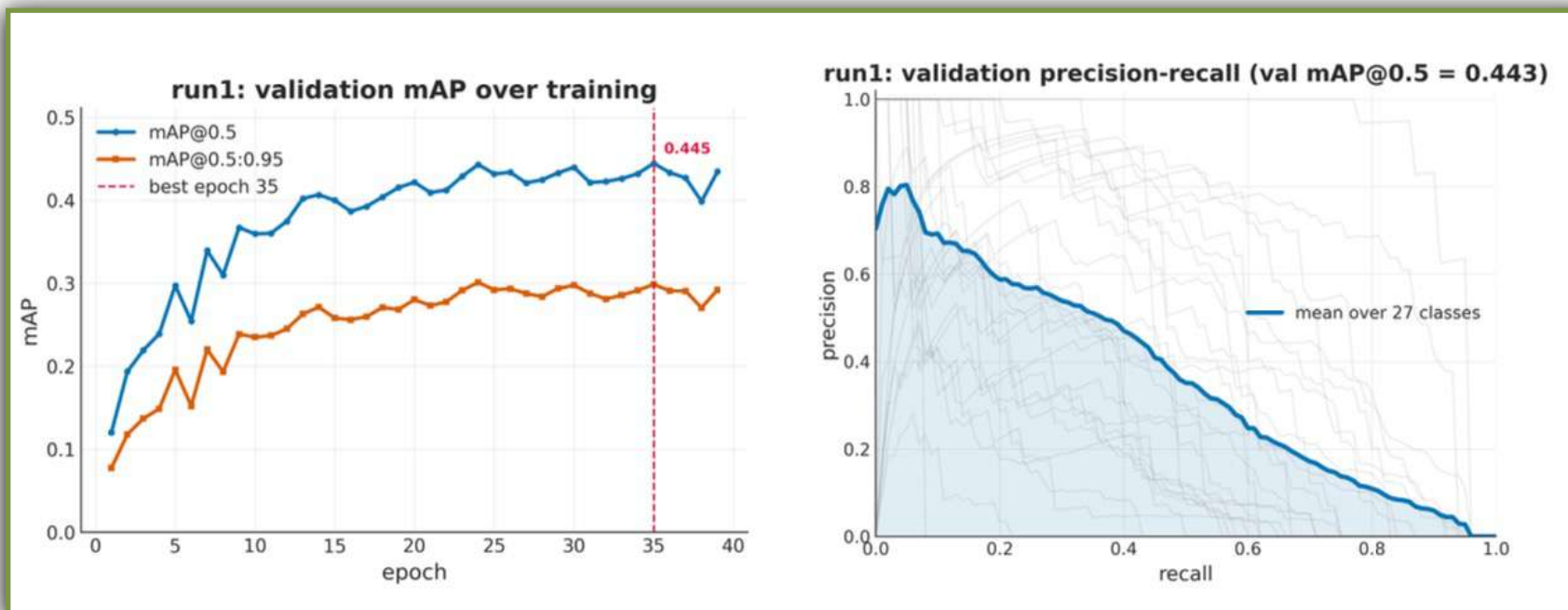


*Flattening of validation loss suggests early stopping to prevent overfitting.*



# Run 1 – Validation

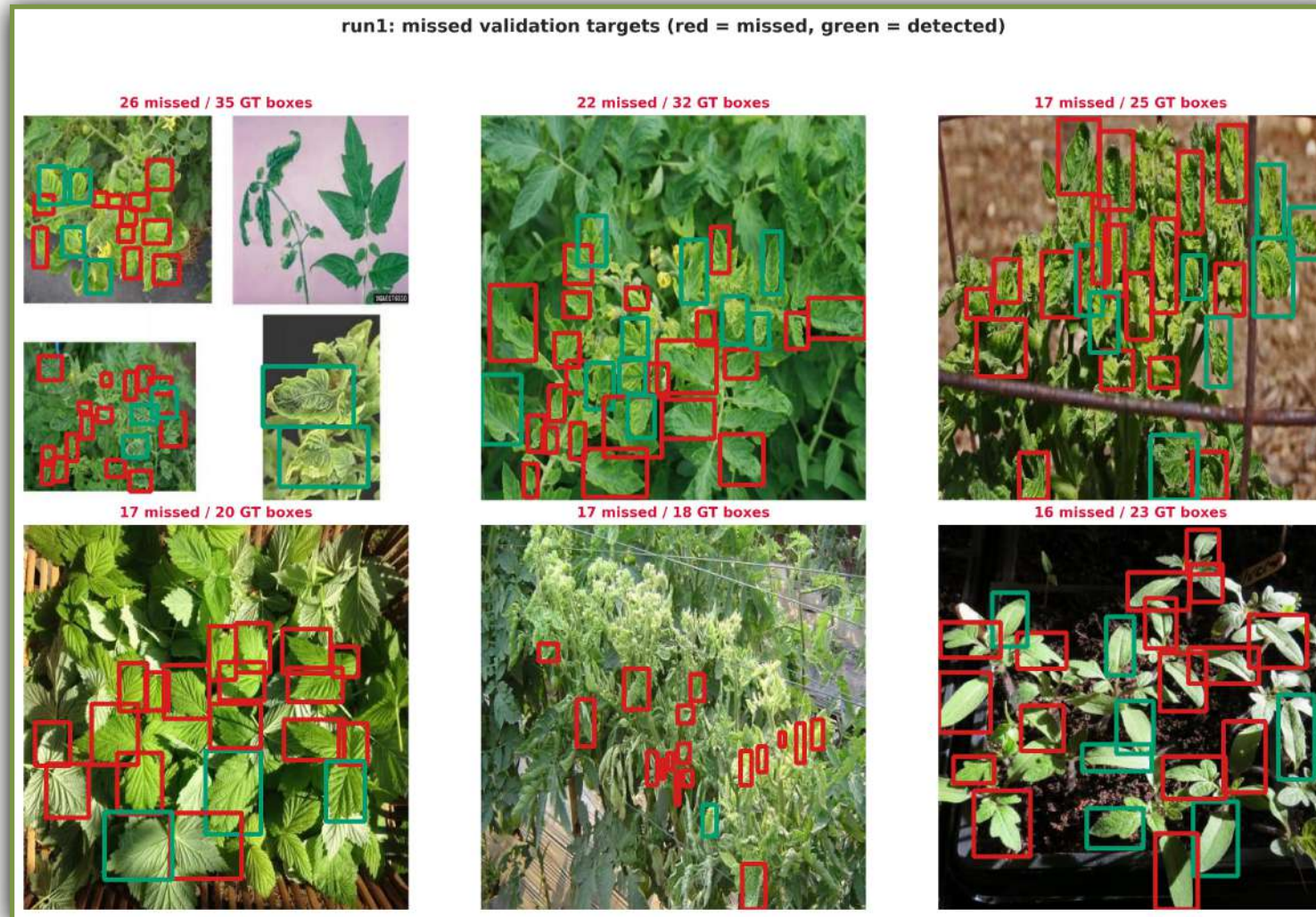
Val mAP@0.5 = **0.443**



*precision gradually decreases at higher recall due to small or rare objects*

# Run 1 – Validation images where Run 1 missed the most objects

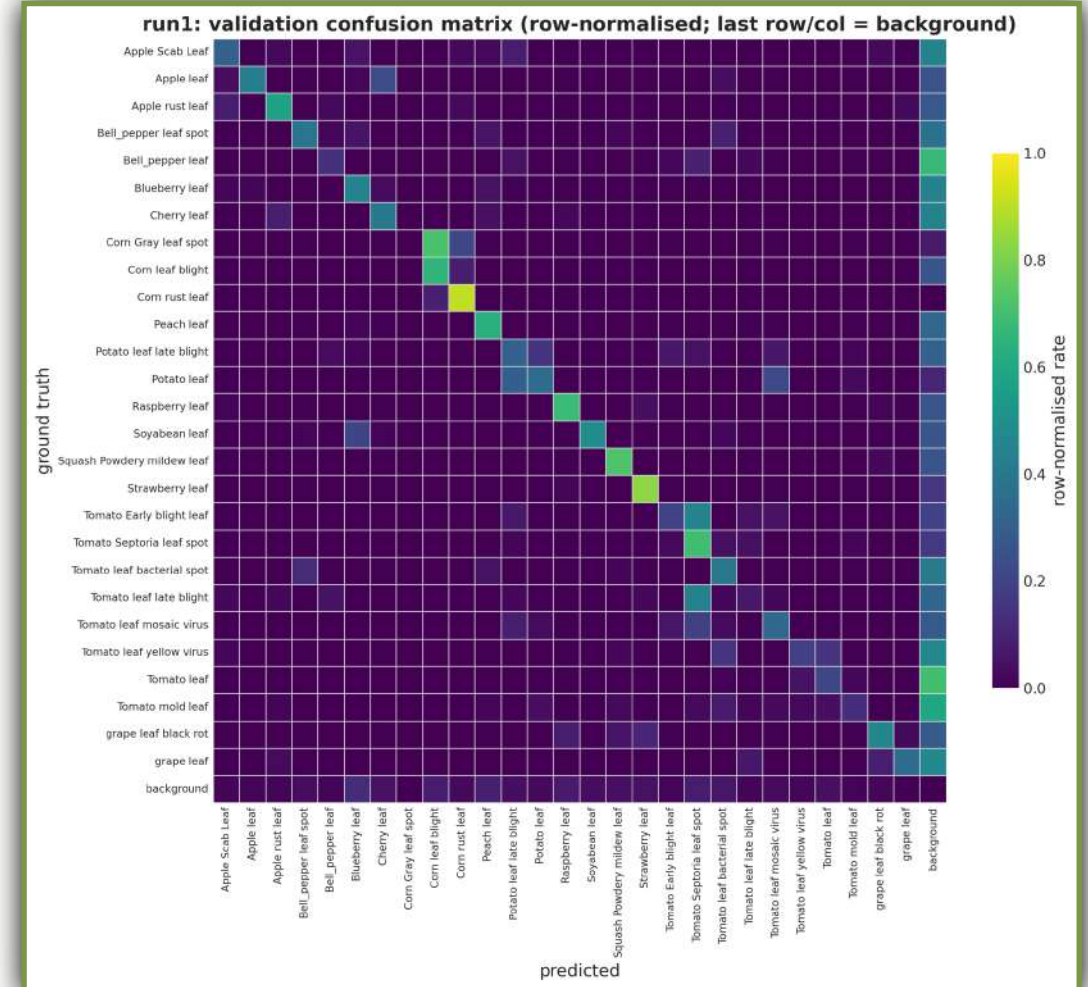
It shows that **small**, **overlapping**, or **dense objects** are the main failure points



# Run 1 – Transfer Baseline

*A clean reference point: what transfer learning alone buys on the cleaned data.*

The diagonal is **correct predictions**. Off-diagonal cells are **class confusions**; the last row & column are **background** — objects missed, or invented from nothing.



## Run 2 – Training

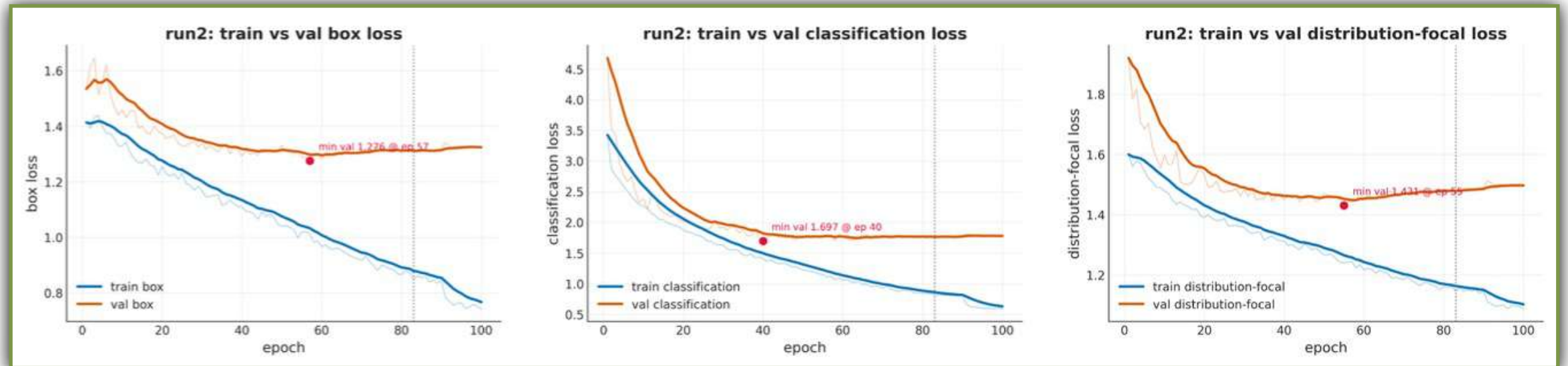
### Configuration

- **Init:** yolov8s.pt (COCO)
- **Optimizer:** AdamW ·  $lr_0$  1e-3 · cosine
- **Schedule:** 100 epochs · patience 20
- **Augmentation** — same values as Run 1

*optimizer + cosine schedule (augmentation unchanged).*



## Run 2 – Training loss



*A smoother, cleaner descent than the baseline.*



## Run 2 – Validation

Val mAP@0.5 = **0.467** — best of the three

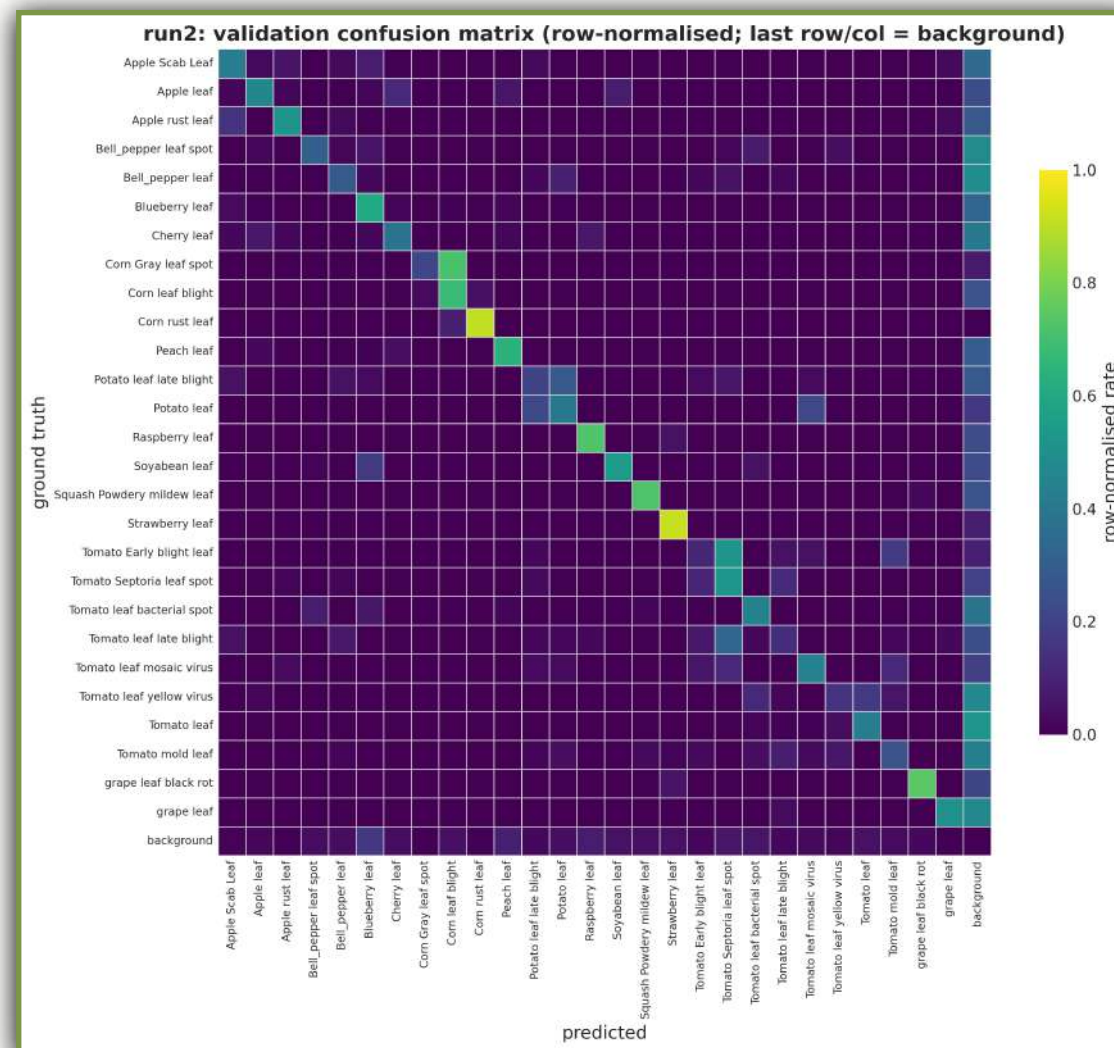


*Surpasses baseline peak in 3 epochs · climbs to 0.374 at epoch 77, still rising.*

## Run 2 – EDA-Tuned Model

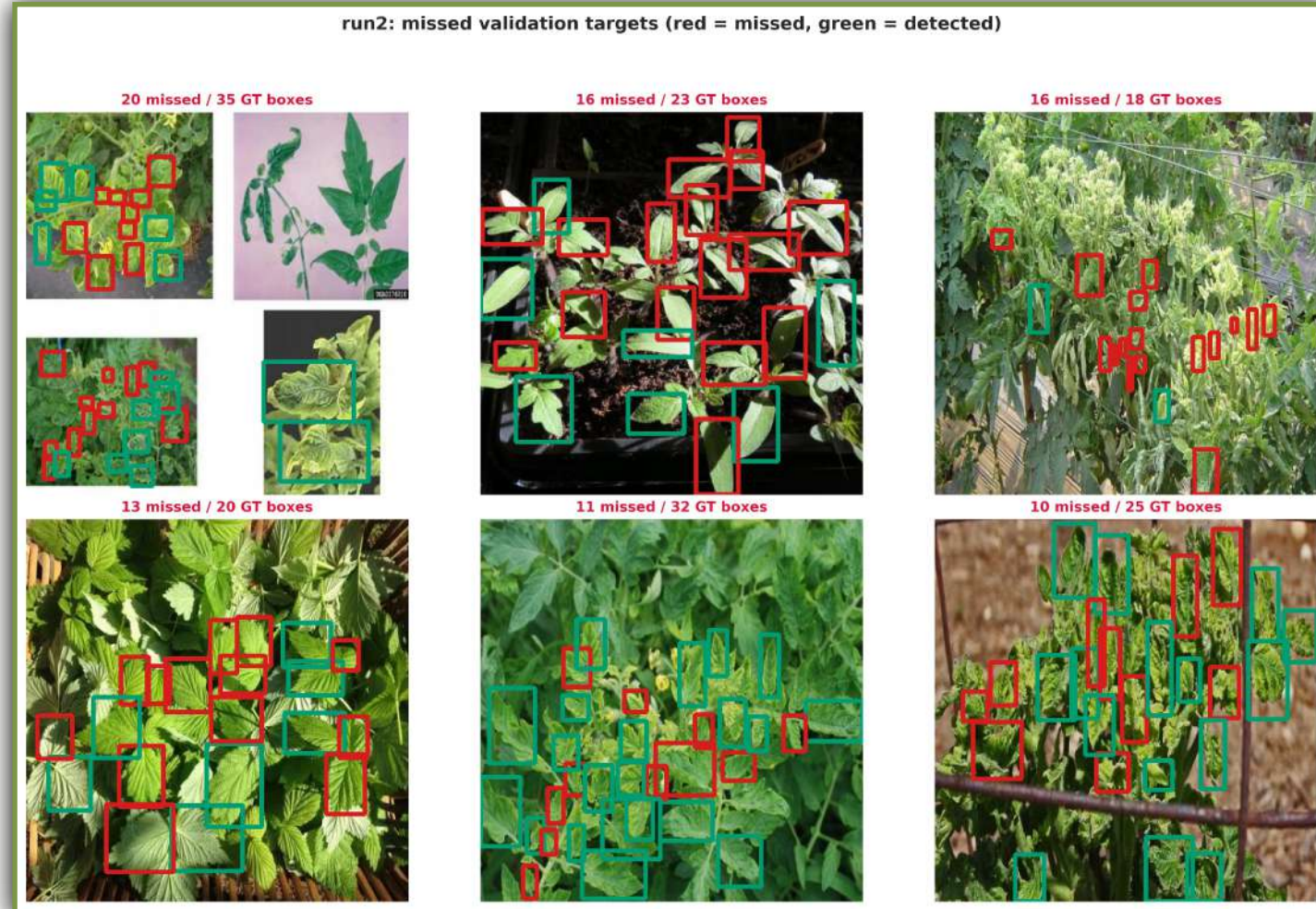
A clean reference point: what transfer learning alone buys on the cleaned data.

Run 2 validation confusion matrix. Compared with Run 1, **more mass sits on the diagonal**, though the look-alike green-leaf classes still trade places



## Run 2 — Validation images where Run 2 missed the most objects

*These Run 2 misses fewer objects overall than Run 1, at the cost of a few more background false positives*

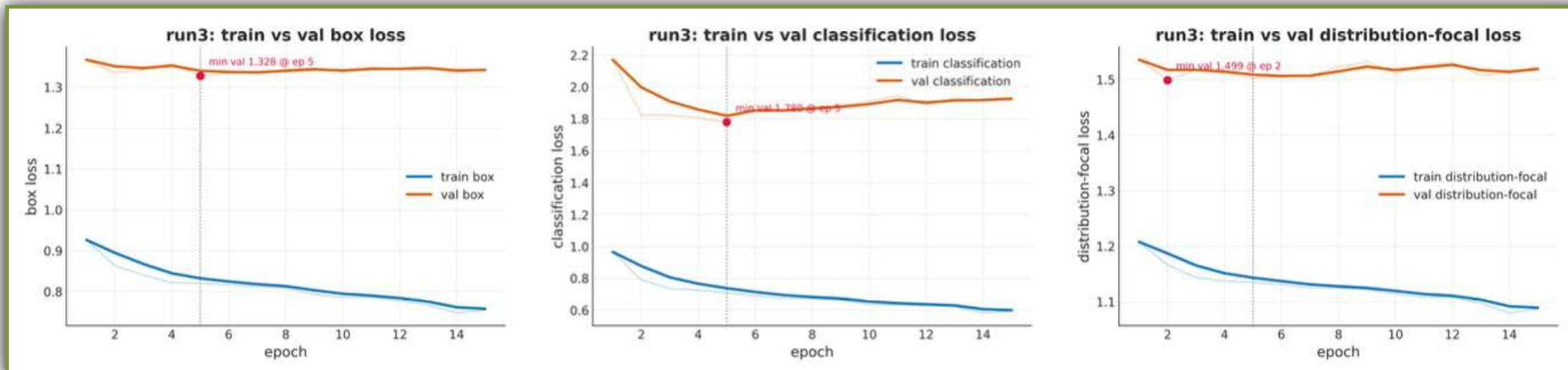


## Run 3 – Training

### Configuration

- Init: Run 2 best.pt (fine-tune)
- Optimizer: AdamW · lr<sub>0</sub> 3e-4 · cosine
- Schedule: 40 epochs · patience 10
- Augmentation — mosaic **OFF** · flip 0.5 · HSV 0.015/0.7/0.4 · eval with TTA

## Run 3 – Training Loss

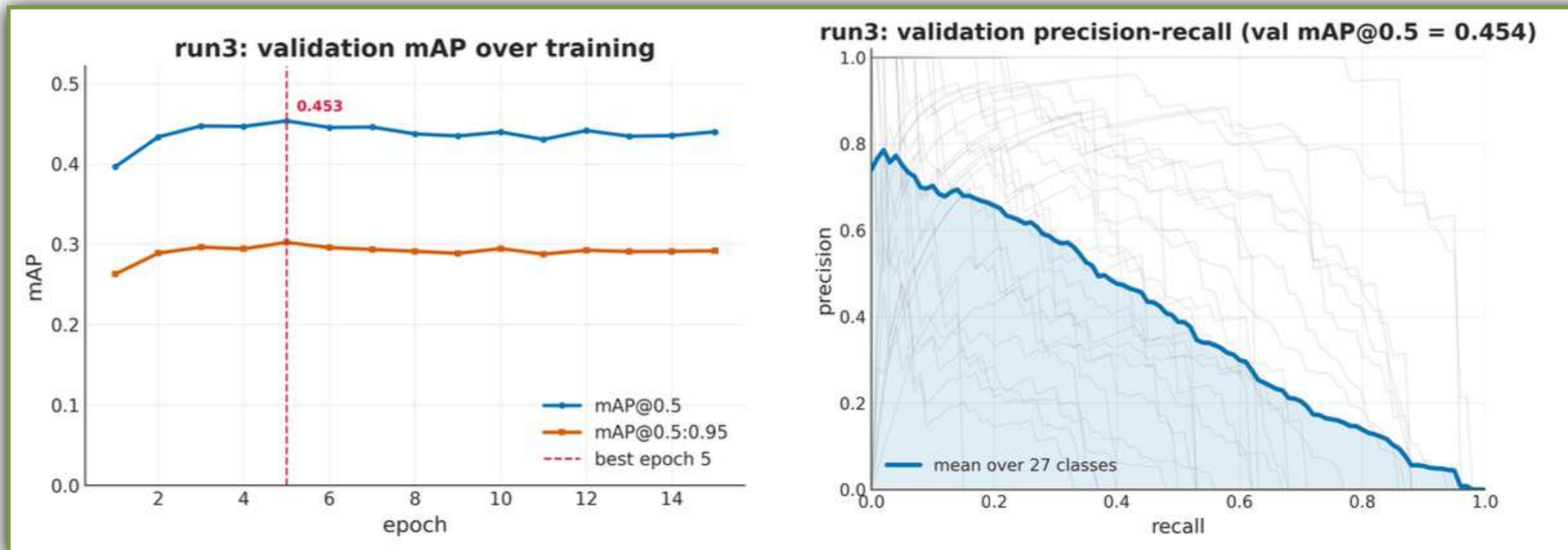


*Lower training loss — but it doesn't translate to validation.*



## Run 3 – Validation

Val mAP@0.5 = **0.454** — below Run 2

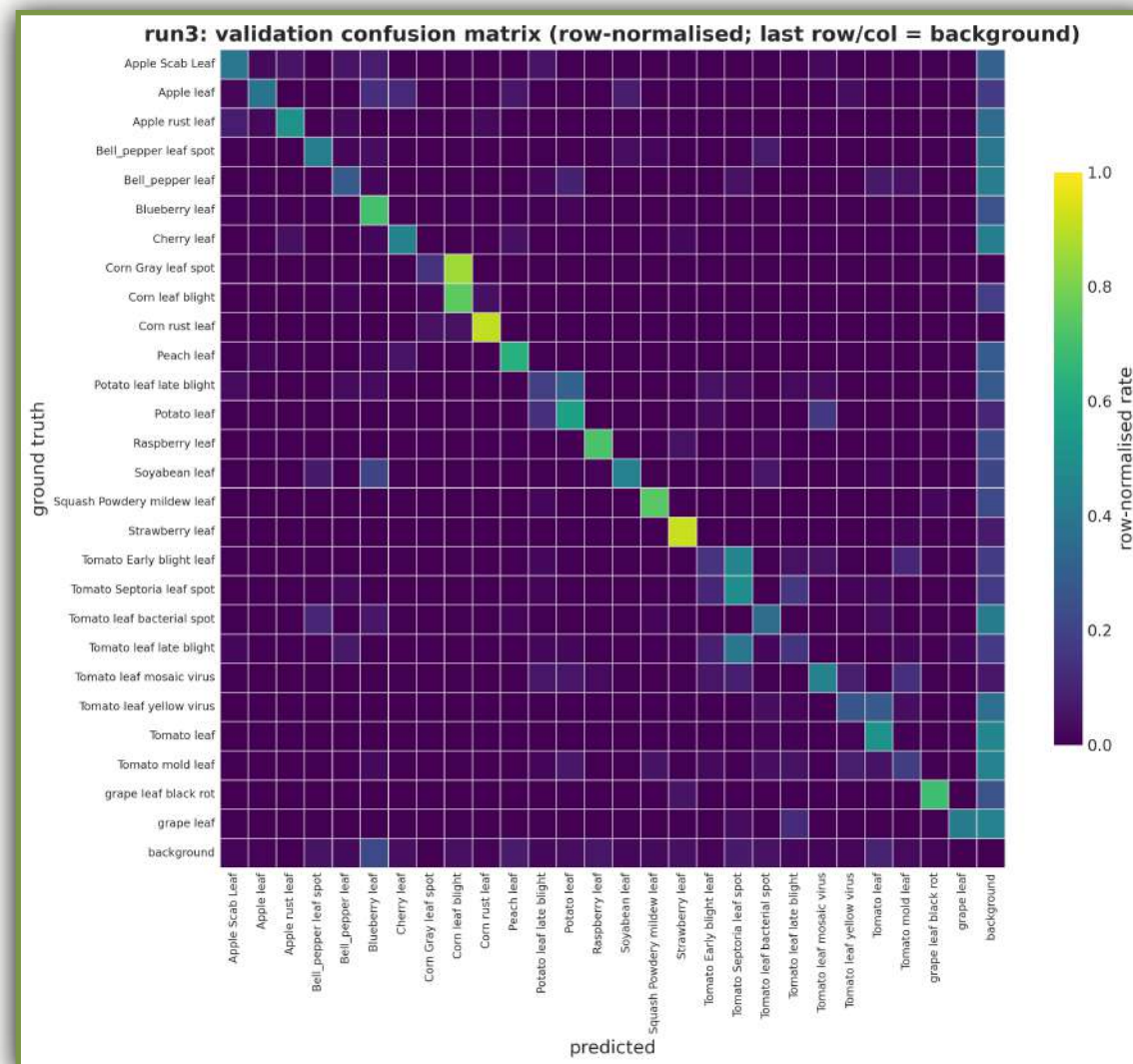


*Run 3 validation curves show stable mAP which may suggest good generalisation*

## Run 3 – Fine-Tuning with TTA

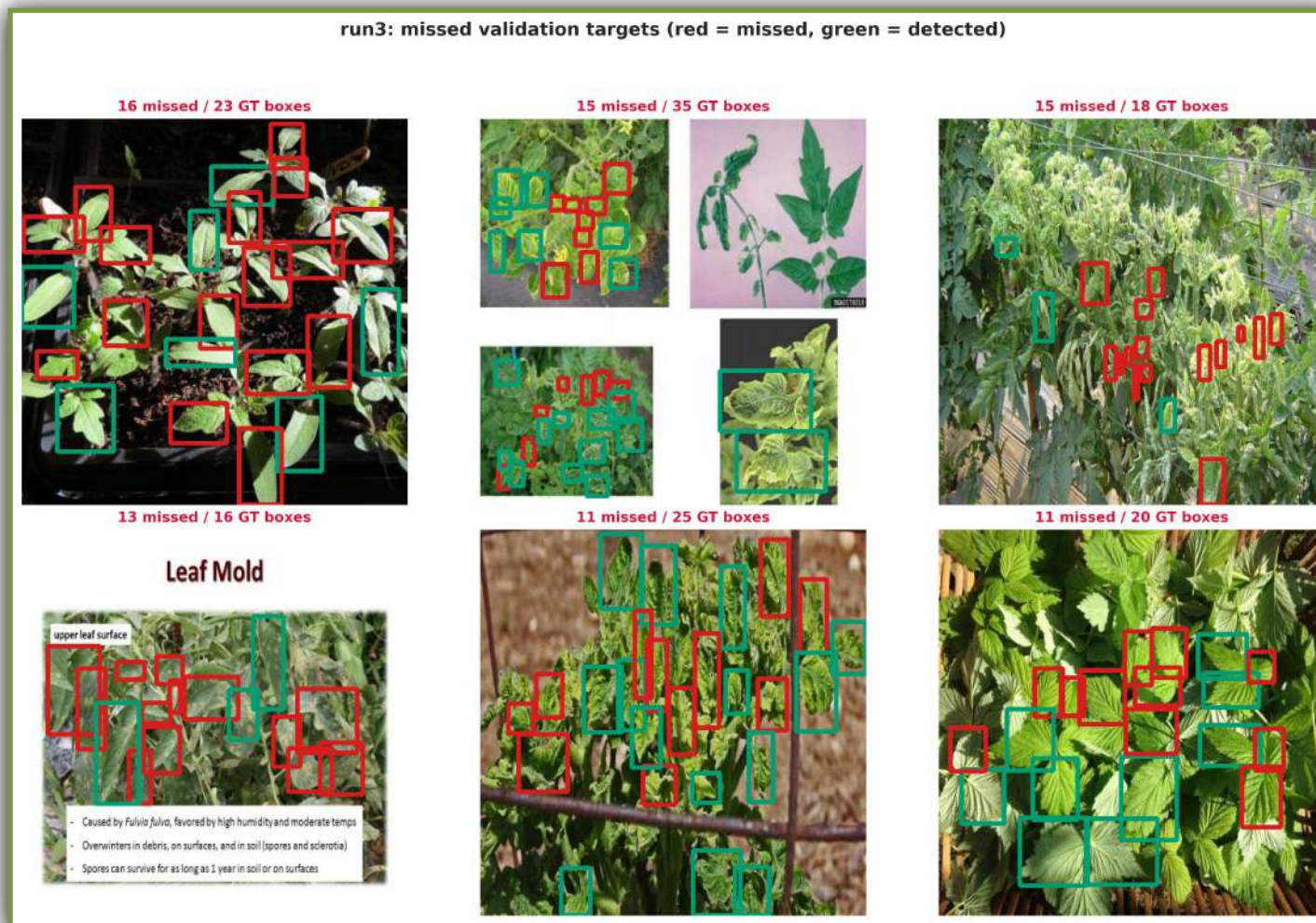
A clean reference point: what transfer learning alone buys on the cleaned data.

Some classes sharpen, but a few – notably Corn Grey leaf spot – degrade under the aggressive fine-tune.



## Run 3 — Validation images where Run 3 missed the most objects

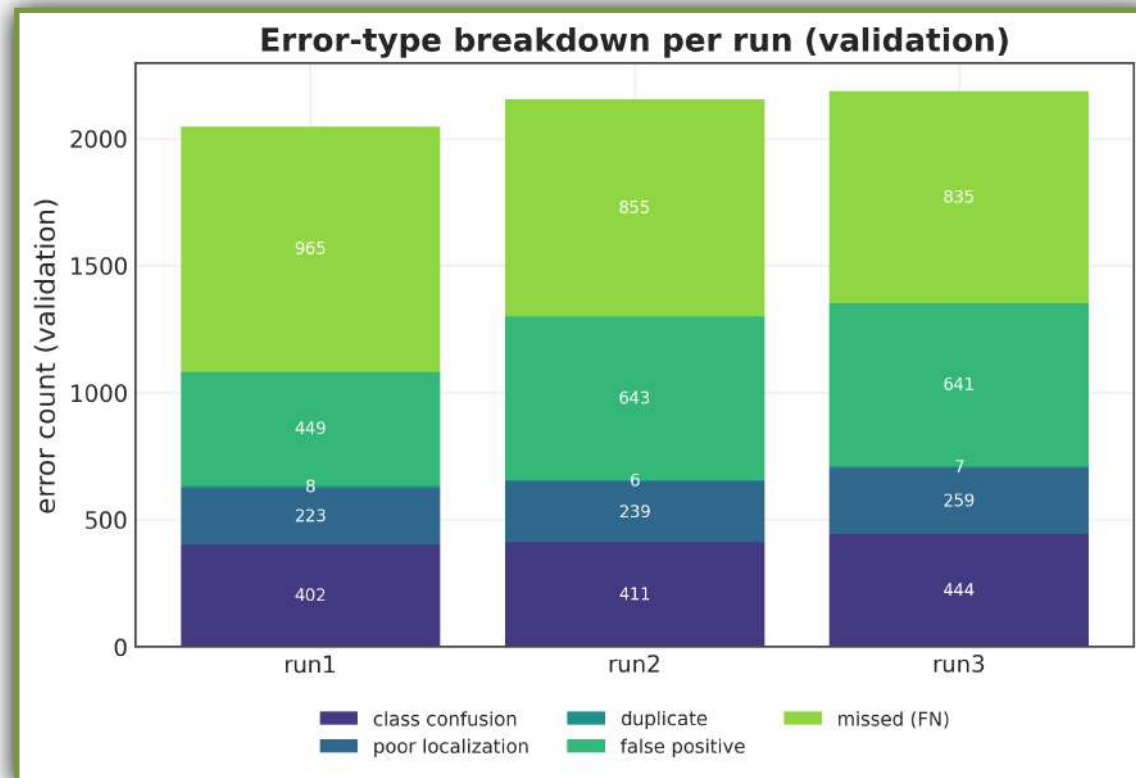
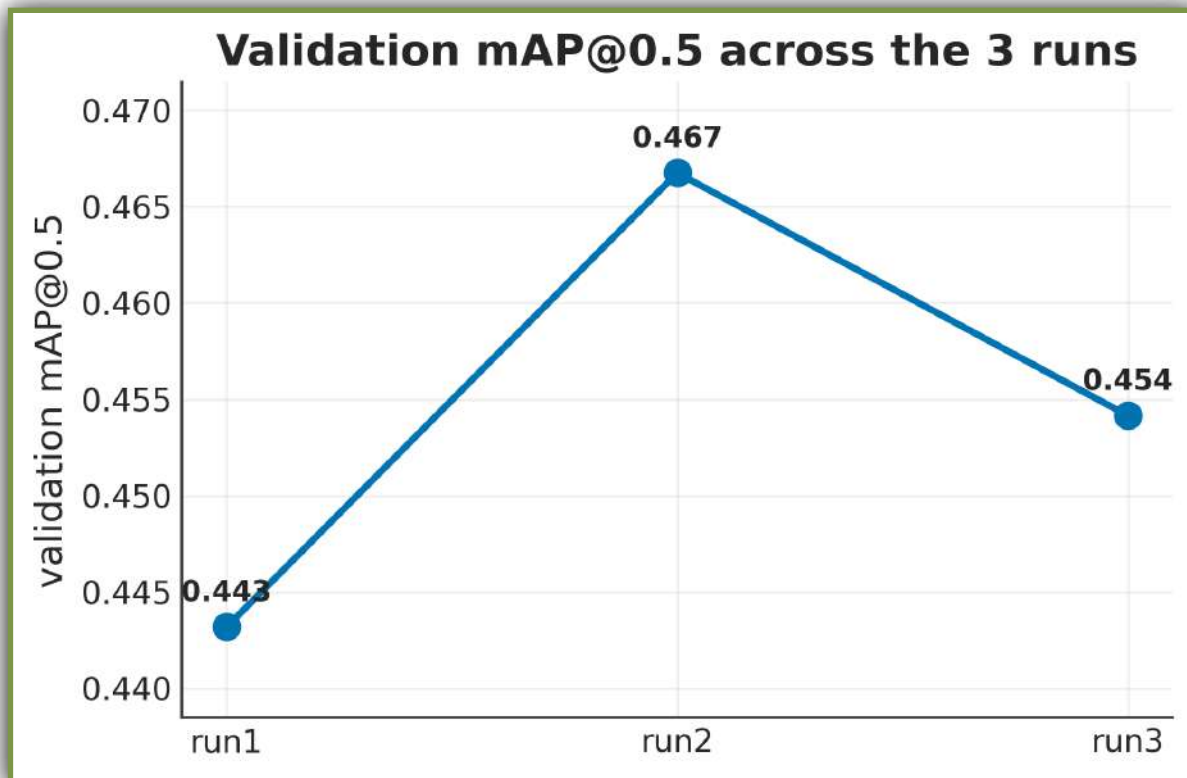
The fine-tune does not reduce misses enough to overtake Run 2.



## COMPARISON

# The three runs, compared

Highest validation mAP — chosen before the test set is opened. Run 2 misses fewer objects — **855 vs 965**.



Selected model → Run 2



# Final Test Evaluation

Run 2, scored once on the 239-image test set including 452 boxes or objects .

| Metric                | Value        |
|-----------------------|--------------|
| mAP@0.5               | <b>0.635</b> |
| mAP@0.50.95           | 0.492        |
| Mean precision        | 0.606        |
| Mean recall           | 0.591        |
| Mean per-class AP@0.5 | 0.635        |

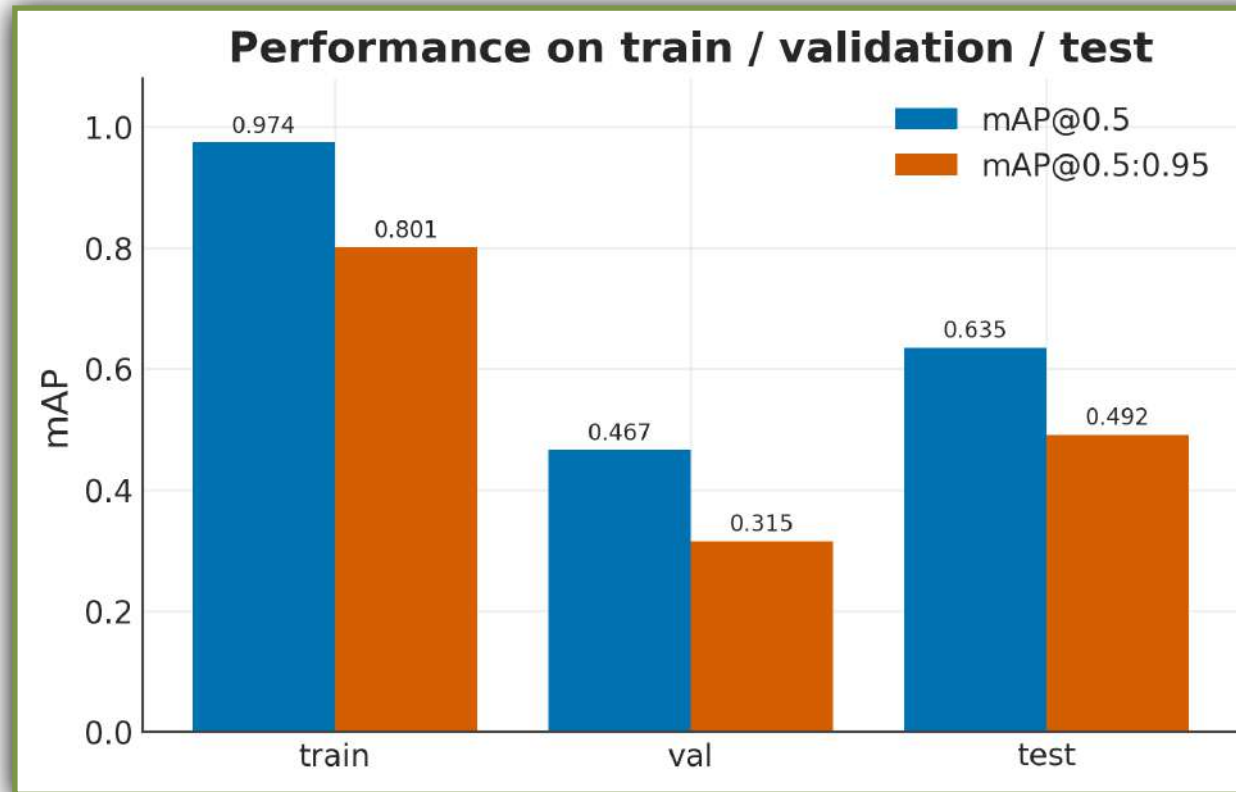
| OUTCOME (OF 454 GT)  | COUNT | %     |
|----------------------|-------|-------|
| True Positives       | 253   | 55.7% |
| Wrong class          | 129   | 28.4% |
| Missed (FN)          | 72    | 15.9% |
| False positives (FP) | 152   | —     |

*The gap between validation and test set is huge !*



# Final Test Evaluation – Is it overfitting?

Yes — and honestly so.

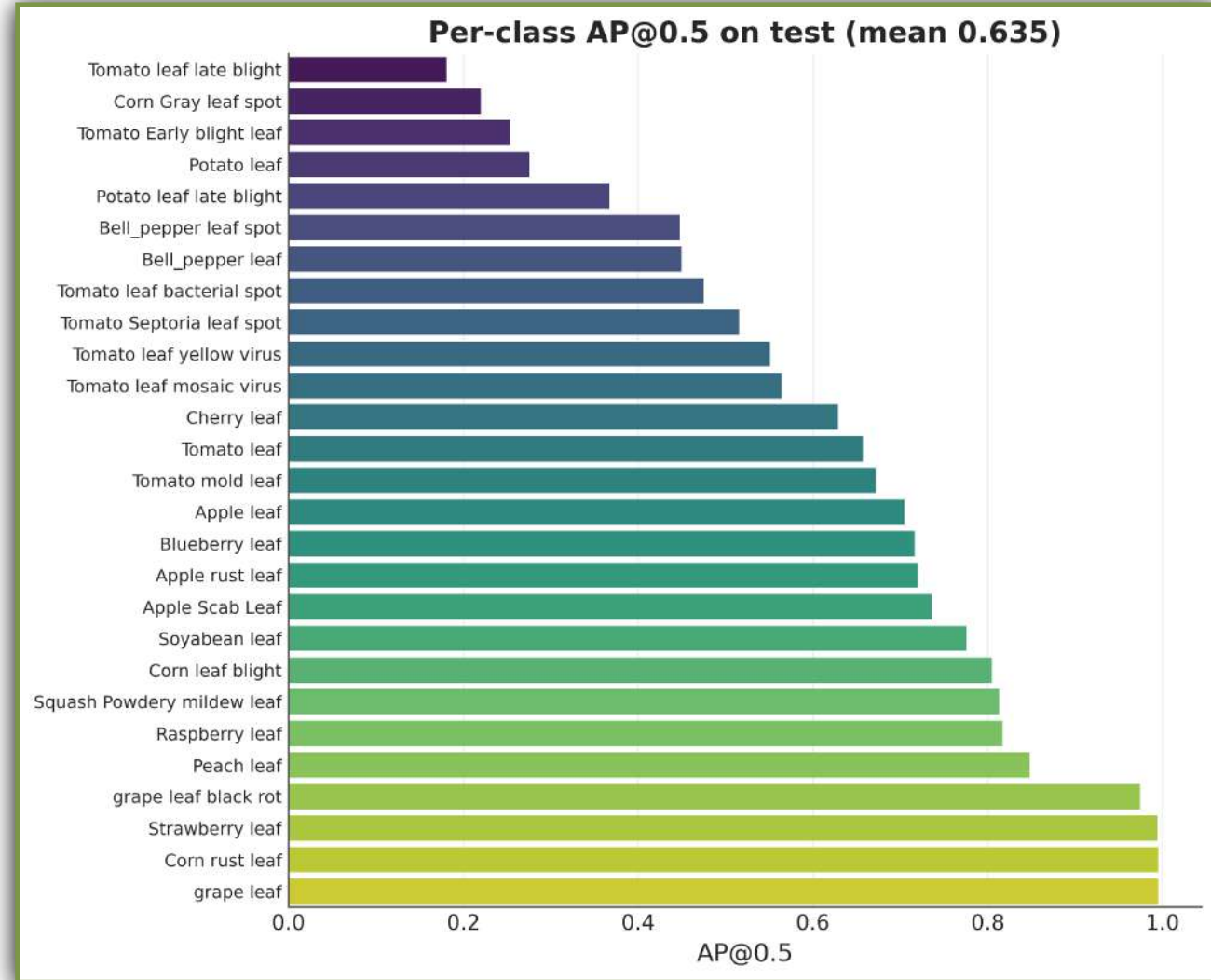


*The gap between training and test/val set is huge !*

## Per-class — strong vs. weak

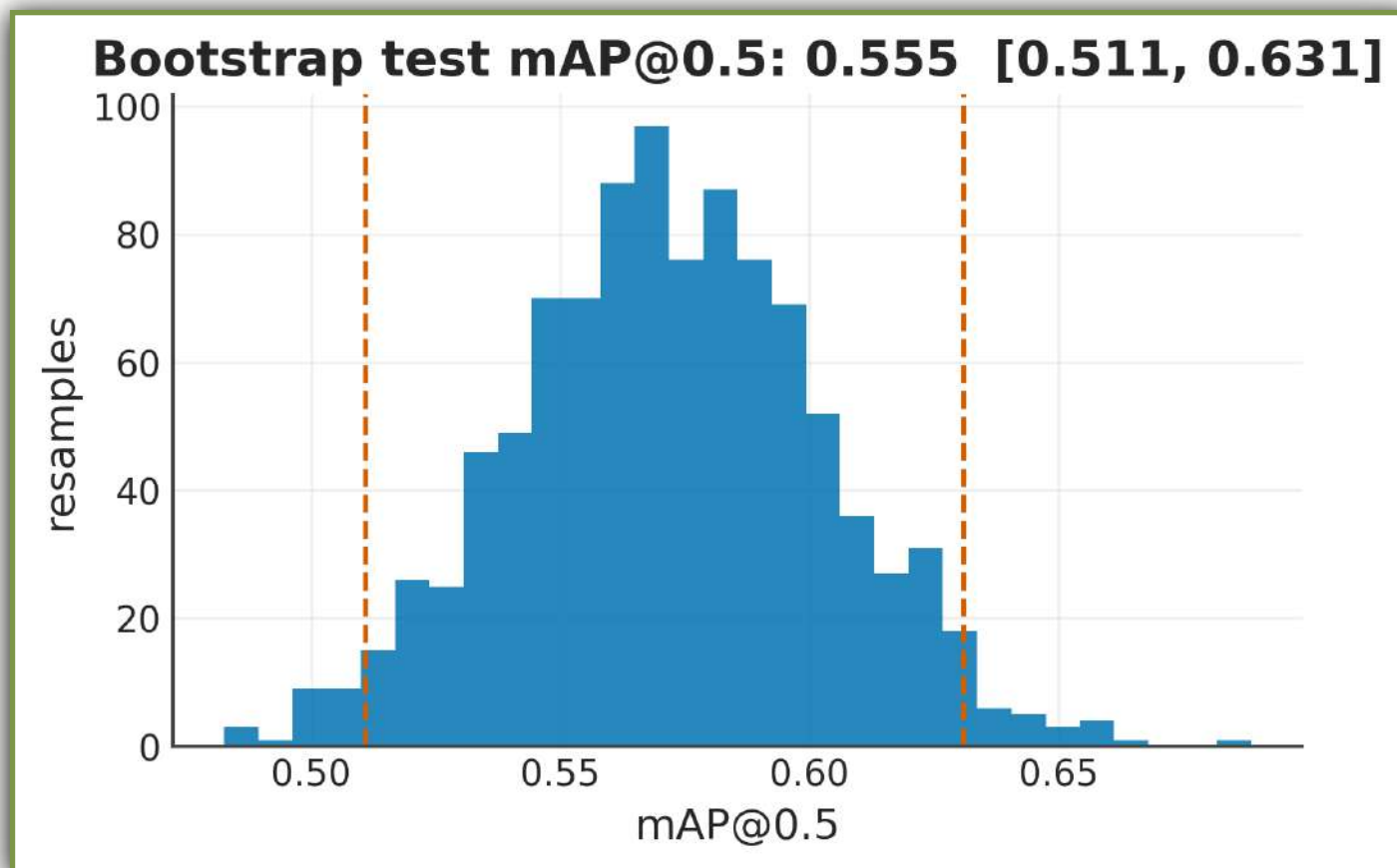
Run 2, scored once on the 239-image test set including 452 boxes or objects .

The strongest classes are **distinctive whole leaf types**; the weakest are **fine-grained tomato/potato disease states** that overlap in colour and texture.



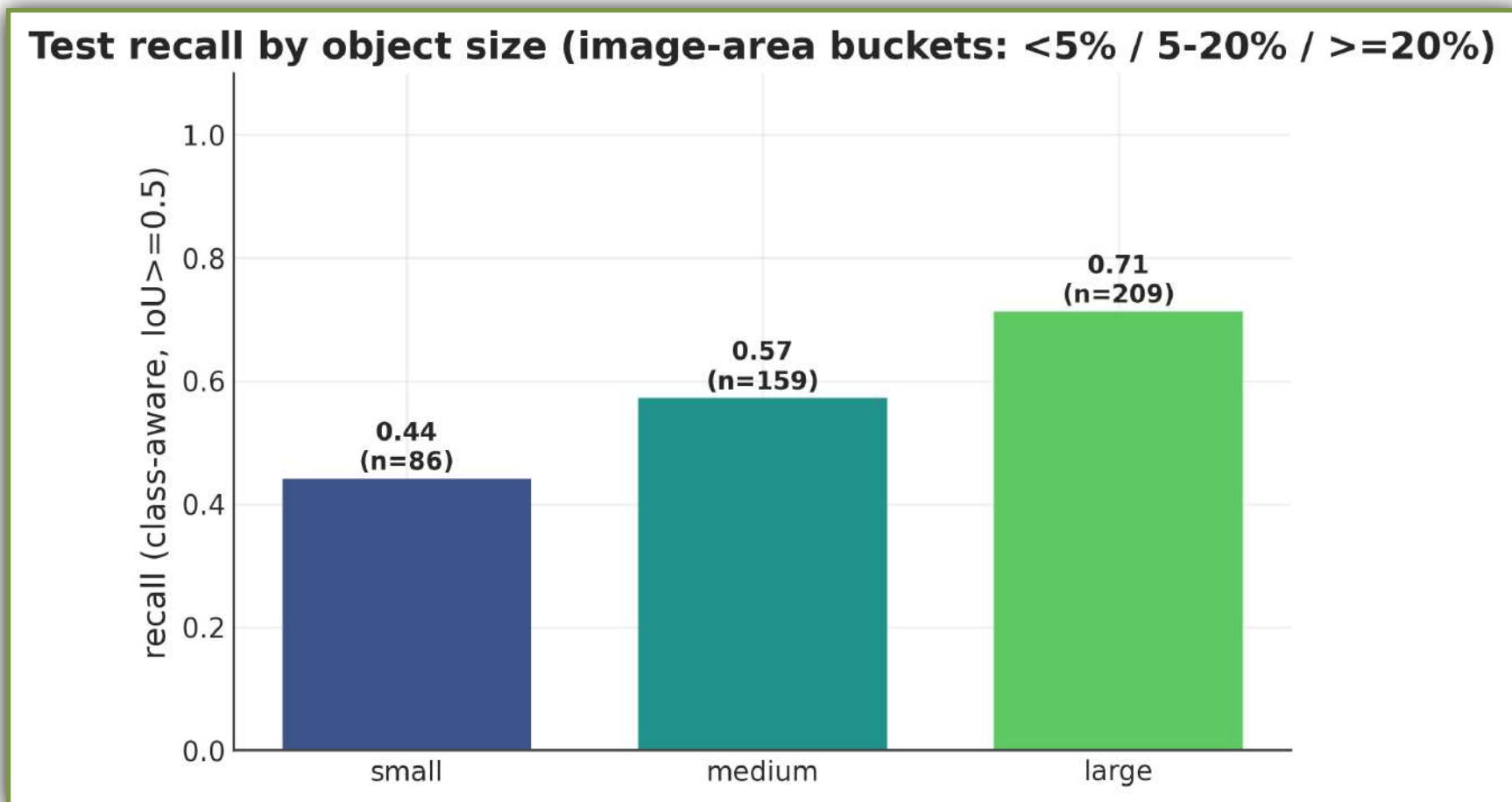
## How sure are we?

95% bootstrap CI [\[0.51, 0.63\]](#) — the honest width of a 239-image test set.



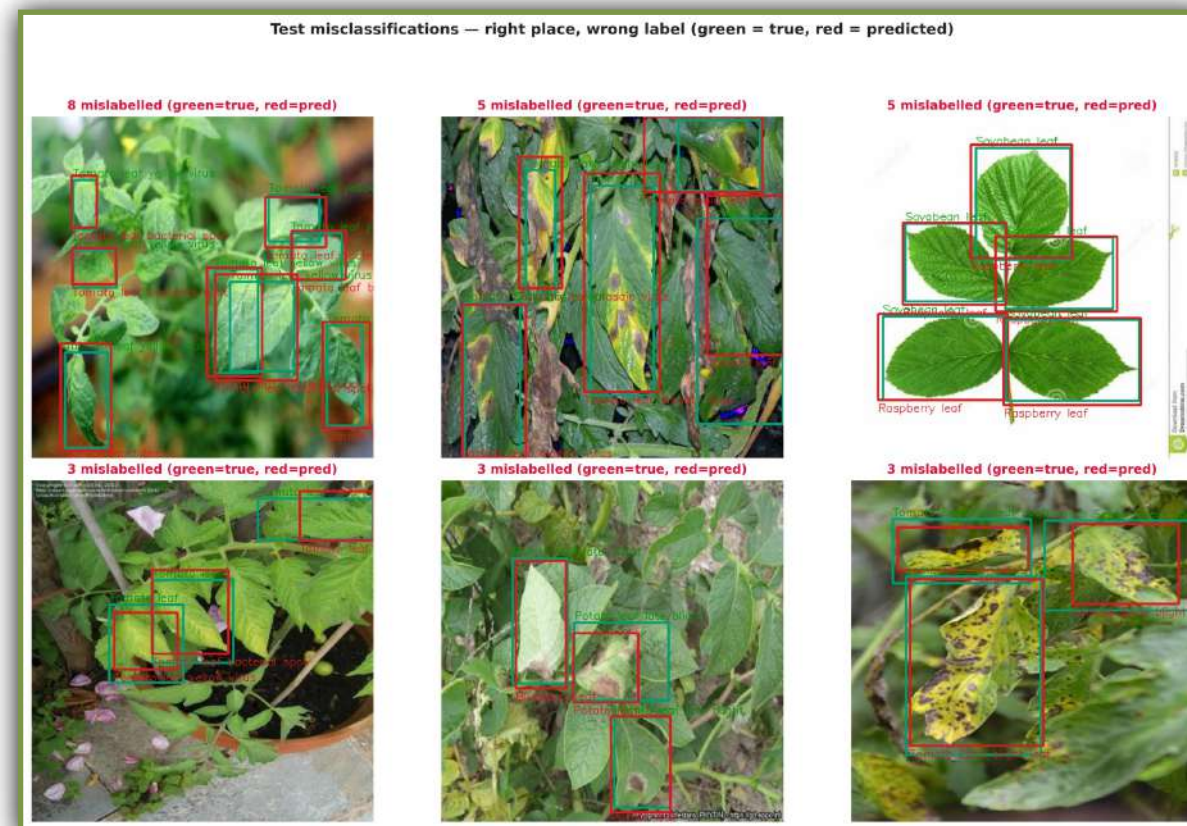
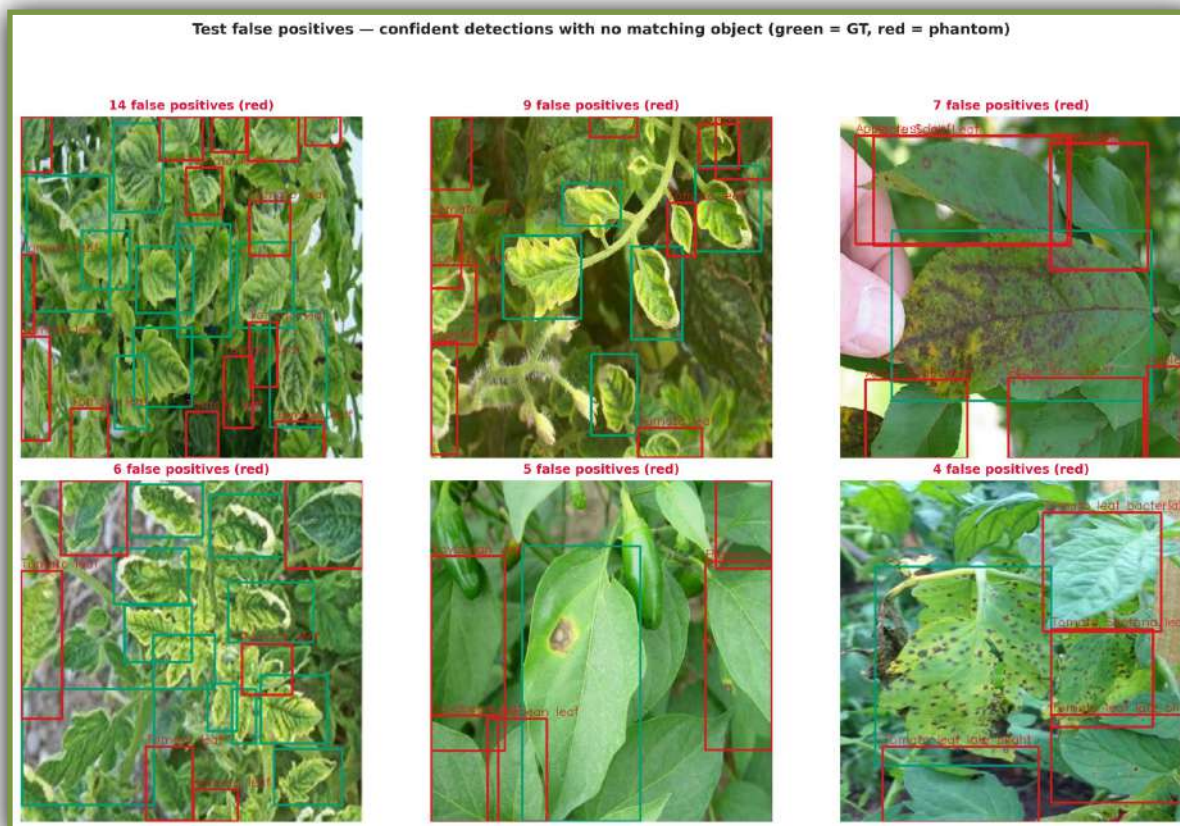
# Does object size matter?

Recall climbs with size



# What do the mistakes look like?

The look-alike confusions the colour analysis flagged in Phase 2, caught on test.







# *Thank you.*

Our model is live here in the bellow link , please give it a try .



<https://leaf-doctor-27classes.streamlit.app/>